



Degree Project in Computer Science and Engineering

Second cycle, 30 credits

Adaptive Hybrid RAG

Optimizing the Efficiency-Accuracy Trade-off via Learned Query
Routing

DANTE WESSLUND

Adaptive Hybrid RAG

Optimizing the Efficiency-Accuracy Trade-off via Learned Query Routing

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Master's Programme, Machine Learning, 120 credits

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Retrieval-Augmented Generation, Graph RAG, Query routing, LightRAG, 2WikiMultihopQA, LLM-as-a-Judge

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Retrieval-Augmented Generation, grafbaserad RAG, frågeroutning, LightRAG, 2WikiMultihopQA, LLM-som-domare

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Acknowledgments

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Stockholm, May 2026

Dante Wesslund

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List of acronyms and abbreviations

API	Application Programming Interface
AUPRC	area under the precision-recall curve
BERT	Bidirectional Encoder Representations from Transformers
DiVA	Digitala Vetenskapliga Arkivet
GPU	Graphics Processing Unit
IMRAD	Introduction, Methods, Results, and Discussion
JSON	JavaScript Object Notation
KI	Karolinska Institutet
KOPPS	Kurs- och programplaneringssystemet
KTH	KTH Royal Institute of Technology
LADOK	Lokalt adb-baserat dokumentationssystem
LLM	large language model
LLM-as-a-Judge	large language model as a judge
QA	question answering
RAG	Retrieval-Augmented Generation
SDG	Sustainable Development Goal
TF-IDF	term frequency-inverse document frequency
TIMTM	Interactive Media Technology
TMMTM	Media Management
UN	United Nations

The list of acronyms and abbreviations should be in alphabetical order based on the spelling of the acronym or abbreviation.

Chapter 1

Introduction

This chapter introduces the thesis topic and motivates the problem studied in the project. Section 1.1 gives the background to retrieval-augmented question answering and the cost of different retrieval modes. Section 1.2 defines the problem and research question. Section 1.3 states the purpose of the work, Section 1.4 presents the goals, Section 1.5 outlines the research methodology, Section 1.6 defines the delimitations, and Section 1.7 describes the structure of the rest of the thesis.

1.1 Background

RAG has become a common architecture for grounding LLM outputs in external documents, especially when the relevant information is private, domain-specific, or changes after model pretraining [1]. A typical RAG system retrieves text passages from a corpus and conditions the generator on those passages, making the language model less dependent on memorized knowledge.

The simplest retrieval layer is vector similarity search over document chunks. It is efficient and often effective for local factual questions, but it can struggle when a query depends on linking evidence across documents, resolving relationships between entities, or comparing distributed attributes. Graph-enhanced RAG systems address this by introducing structured representations of entities, relations, or higher-level document organization. Recent systems such as GraphRAG and LightRAG show how graph-based structure can support retrieval beyond isolated nearest-neighbor chunks [2, 3].

Richer retrieval is not free, though. Graph construction, graph traversal,

larger context assembly, and additional language-model calls can increase execution time. At Ericsson, where internal technical repositories are large and fragmented, graph-enhanced retrieval may improve answer quality for difficult questions, but applying it to every query can waste computation on questions that cheaper vector retrieval already handles [4, 5].

This tension motivates adaptive retrieval. Self-RAG and Adaptive-RAG argue that retrieval behavior should depend on the query rather than follow a fixed policy [6, 4]. More broadly, the model-routing literature frames this as a cost–performance decision, where different execution paths carry different expected benefits and costs, and a router should select the path appropriate for the input [5].

The question studied here is narrower than general model routing. Instead of routing between different language models, the project routes between retrieval modes inside a fixed Hybrid RAG setting. The low-cost path is LightRAG’s *Naive* mode, used as a vector-retrieval baseline, and the high-cost path is LightRAG’s *Mix* mode, which combines vector retrieval with graph-enhanced retrieval. The routing layer is intentionally lightweight, because if the router itself requires a large decoder model, much of the intended efficiency gain is lost. Prior work on encoder-only classification supports this design for narrow text-classification tasks [7].

1.2 Problem

Hybrid RAG systems expose a practical tension between retrieval quality and retrieval cost. A cheap vector-based path suffices for many questions, while a graph-enhanced path helps when the answer depends on relationships or evidence distributed across several documents. A static policy cannot exploit this distinction, because always using the cheap path risks missing cases where structured retrieval is needed, while always using the expensive path wastes computation on easier queries.

The project focuses on that decision point. In LightRAG, the choice is operationalized as *Naive* retrieval (the lower-cost vector path) versus *Mix* retrieval (the more expensive graph-enhanced path). Prior work motivates both graph-based RAG and adaptive retrieval [2, 3, 6, 4], but does not answer whether this particular retrieval-mode choice can be predicted before retrieval, from the query text alone.

1.2.1 Original problem and definition

The practical motivation comes from industrial QA over large technical repositories. Ericsson works with internal documentation where relevant information is fragmented across documents and absent from public language-model pretraining. RAG is needed for grounding, but graph-enhanced retrieval should ideally be reserved for queries where the cheaper vector path is insufficient.

The engineering problem is to decide, for a given query, whether cheap vector retrieval suffices or whether graph-enhanced retrieval is required, and to make that decision before the expensive path runs. A ModernBERT-based query classifier serves this role instead of a large general-purpose LLM router, keeping the control layer itself cheap [7, 8].

1.2.2 Scientific and engineering issues

The scientific issue is whether retrieval need is predictable at all from the query alone. If the distinction between vector-sufficient and graph-required queries is mainly determined by corpus state after retrieval, a query-only router should not work. If the distinction is partly reflected in how the question is phrased, a supervised classifier may learn a useful decision boundary.

The engineering issue is that 2WikiMultihopQA provides answer supervision but not routing labels directly. The project therefore constructs routing labels by executing both retrieval modes on 2WikiMultihopQA and comparing the outputs against benchmark answers using LLM-as-a-Judge, treating the judge as a fallible labeling instrument rather than unquestioned ground truth [9, 10].

1.2.2.1 Research question

Within a fixed Hybrid RAG setting, how predictable is the distinction between queries for which vector retrieval is sufficient and queries for which graph-enhanced retrieval is required, based on the query text alone?

1.3 Purpose

The purpose is to investigate whether retrieval-mode selection in a Hybrid RAG system can be made adaptive without moving the routing decision into another expensive model call. Graph-enhanced retrieval should be used when it is needed, but avoided when vector retrieval already suffices.

For Ericsson, internal technical documentation is large, fragmented, and absent from public language-model pretraining. High-quality grounded QA is useful, but indiscriminately applying the most expensive retrieval strategy creates unnecessary latency. A successful router would reduce redundant graph-enhanced retrieval while preserving access to it for queries that genuinely require structured evidence.

On the academic side, the goal is narrower than it might seem. The project does not aim to prove that graph retrieval is universally superior, nor that LightRAG is the best possible RAG framework. The question is whether the need for graph-enhanced retrieval is predictable enough from the query text that a lightweight classifier can serve as a useful control layer. The contribution is therefore empirical, asking whether the vector-sufficient versus graph-required distinction can be learned in a fixed experimental setting.

1.4 Goals

The main goal is to design and evaluate a learned routing layer that predicts whether a query should use LightRAG’s *Naive* or *Mix* retrieval mode. Four concrete objectives structure the work.

1. Implement a reproducible pipeline for preparing 2WikiMultihopQA samples, indexing them in LightRAG, and executing both *Naive* and *Mix* retrieval modes.
2. Construct a supervised routing dataset by labeling each query according to the minimum viable retrieval mode, using LLM-as-a-Judge to compare retrieval outputs against benchmark answers.
3. Train lightweight query-only routers, including a ModernBERT-based classifier and a classical TF-IDF logistic-regression baseline, to predict whether graph-enhanced retrieval is required.
4. Evaluate the routers both as held-out classifiers using metrics such as AUPRC and as thresholded routing policies through offline simulation against majority-class, always-*Naive*, and always-*Mix* baselines.

1.5 Research Methodology

The study is a quantitative empirical experiment on the public 2WikiMultihopQA benchmark. LightRAG serves as a fixed Hybrid RAG system;

every query is run in both *Naive* and *Mix* retrieval modes. The generated answers are then evaluated against benchmark answers with an **LLM-as-a-Judge** procedure, producing routing labels that indicate whether vector retrieval was sufficient or graph-enhanced retrieval was required.

These labels are used to train lightweight query-only routers, a ModernBERT-based classifier and a **TF-IDF** logistic-regression baseline. Evaluation on held-out data uses classification metrics such as **AUPRC**, precision, recall, and F1 to measure how reliably the router predicts the judged routing decision.

To connect classifier performance to system efficiency, the predicted probabilities are also evaluated as routing policies. Different thresholds are compared against static baselines (always *Naive*, always *Mix*) using measured execution times and judged answer outcomes.

1.6 Delimitations

The scope is deliberately limited to query-only retrieval-mode routing. The router receives the query text as input and does not inspect retrieved documents, intermediate graph state, answer drafts, or model uncertainty after generation. This makes the routing problem cheaper and cleaner, but it means the project does not evaluate routers that adapt after retrieval has already begun.

The retrieval setting is LightRAG with two modes: *Naive* and *Mix*. These operationalize the broader vector-versus-graph distinction, but results should not be assumed to transfer automatically to every Hybrid **RAG** framework.

The primary benchmark is 2WikiMultihopQA [9], chosen because it is public, provides supporting evidence, and makes the experiment reproducible without depending on confidential data. Ericsson documentation motivates the industrial problem but is not the core benchmark.

Routing labels come from **LLM-as-a-Judge** and are treated as judged supervision, not absolute ground truth. Cases where neither retrieval mode produces a correct answer are excluded from router training.

1.7 Structure of the thesis

Chapter 2 presents the technical and scientific background on **RAG**, graph-enhanced retrieval, adaptive retrieval, query routing, ModernBERT, and related work. Chapter 3 describes the experimental method, including data preparation, LightRAG execution, label generation, model training, and

evaluation design. Chapter 4 describes the implemented pipeline and router models. Chapter 5 presents the experimental results and analyzes the routing policies. Chapter 6 discusses the implications and limitations of the results. Chapter 7 concludes with answers to the research question and suggestions for future work.

Chapter 2

Background

This chapter provides the theoretical and related-work basis for the thesis. Section 2.1 introduces RAG. Section 2.2 explains the distinction between vector retrieval and graph-enhanced retrieval. Section 2.3 describes adaptive retrieval and query routing. Section 2.4 positions the thesis in relation to previous work.

2.1 Retrieval-Augmented Generation

RAG connects a generative language model to external knowledge. The original formulation combines parametric knowledge in a neural language model with non-parametric memory retrieved from a document collection [11]. Later surveys describe RAG more broadly as a family of systems for grounding LLM outputs in external documents, databases, or other knowledge sources [1]. For industrial QA the motivation is straightforward: the relevant facts are often private, domain-specific, or updated after the language model was trained.

A simplified RAG process can be written as a retrieval-conditional generation problem, following the retrieval-marginalization view in the original formulation but using notation adapted for this project [11]. Given a query q , a retriever selects a set of documents or chunks $\mathcal{D}_k(q)$, and a generator produces an answer a conditioned on the query and the retrieved context.

$$p(a | q) = \sum_{d \in \mathcal{D}_k(q)} p_\theta(a | q, d) p_\eta(d | q). \quad (2.1)$$

Equation 2.1 is not meant as an implementation claim about every modern RAG system, but it captures the dependency that matters here. The

generated answer depends on which evidence the retrieval system exposes to the language model. Retrieval is not a passive preprocessing step, it is a central part of the system's behavior.

At Ericsson, internal technical documents cannot be assumed to exist in public model pretraining, so a **RAG** system is needed to ground answers in company-controlled knowledge. But the retrieval method must still decide how much structure and computation to use for each query. Grounded **QA** is useful; indiscriminate use of expensive retrieval wastes resources.

2.2 Vector Retrieval, Hybrid Retrieval, and Graph-Based RAG

Most practical **RAG** systems start with vector retrieval. Documents are split into chunks, each chunk is embedded into a vector space, and queries are matched to chunks by similarity. The original **RAG** model indexes chunked Wikipedia passages and retrieves top- k passages through maximum inner-product search [11]. Dense Passage Retrieval takes a similar approach, representing questions and passages with dense encoders and retrieving by vector similarity for open-domain **QA** [12]. Vector retrieval is therefore a natural low-cost baseline.

The weakness of pure vector retrieval is that similarity to individual chunks does not always capture relational structure. Multi-hop **QA** benchmarks are built around questions whose answers require multiple reasoning steps or supporting evidence from more than one location [13, 9, 14]. If the required evidence is split across documents, a nearest-neighbor chunk search may retrieve only part of the information needed to answer correctly.

Graph-enhanced **RAG** systems address this limitation by adding structured representations of entities, relations, or communities. GraphRAG, for example, builds graph-based structure over a corpus and uses that structure to support query-focused summarization and reasoning over private document collections [2]. LightRAG similarly aims to combine simple retrieval with graph-enhanced representations while exposing multiple retrieval modes [3]. These systems motivate the idea that richer retrieval can help when the query requires more than local passage similarity.

The cost side matters just as much. Graph-enhanced systems require additional work for graph construction, traversal, context assembly, and language-model calls. The GraphRAG pipeline, for example, uses an **LLM** to build graph structure and generate community-level summaries, and

Microsoft’s own cost discussion identifies repeated LLM calls during graph instruction as a major cost driver [2, 0]. Graph retrieval is therefore not an unconditional replacement for vector retrieval, but an expensive capability that should ideally be used only when it is needed.

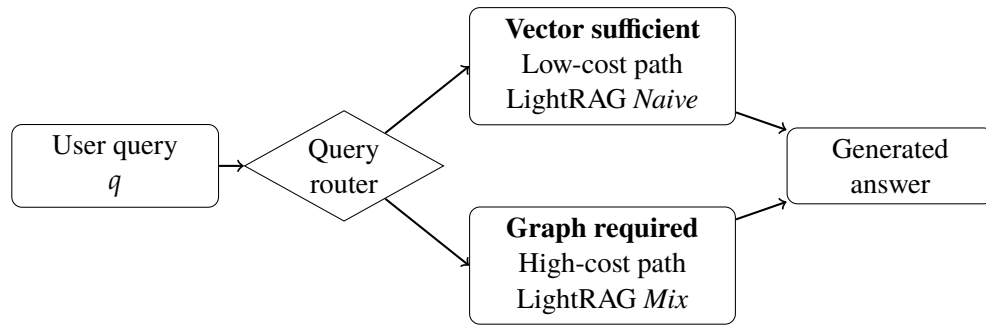


Figure 2.1: The retrieval-mode routing problem studied in this thesis. The router should select a cheap vector-like path when it is sufficient and a graph-enhanced path when the query requires richer retrieval.

2.3 Adaptive Retrieval and Query Routing

Adaptive retrieval challenges the assumption that the same retrieval strategy should be used for every input. Self-RAG learns to retrieve, generate, and critique through self-reflection, making retrieval behavior conditional rather than fixed [6]. Adaptive-RAG studies how question complexity can guide the selection of retrieval-augmented reasoning strategies [4]. Both support the premise that retrieval should depend on the query.

The routing literature frames this more explicitly as a cost–performance problem. RouterBench studies multi-model routing, where different language models have different expected cost and quality characteristics [5]. Although RouterBench concerns model routing rather than retrieval-mode routing, the abstraction carries over directly because a router observes an input and chooses among execution paths before the expensive computation happens.

The move made here is to adapt routing from the model layer to the retrieval layer. Instead of asking which language model should answer, the project asks which retrieval mode should supply evidence. The cost-aware intuition is the same as in RouterBench and the query-dependent motivation is the same as in Adaptive-RAG, but the action space is narrower [5, 4]. With the retrieval framework fixed and only two modes available, the scientific question becomes measurable.

An important constraint is that the router is query-only. It sees the text of the query and predicts whether graph-enhanced retrieval is required before any retrieval runs. This is consistent with adaptive-retrieval work that predicts query complexity before selecting a strategy [4, 5]. It makes the task harder than routing with access to retrieved documents or generated answers, but it preserves the potential efficiency gain. A router that needs to run an expensive retrieval step before deciding would undermine its role as a cheap control layer.

Encoder-only language models are well suited to this kind of narrow classification. BERT introduced deep bidirectional transformer representations for language understanding [0], and later comparisons between encoder-only and decoder-only architectures confirm that encoders remain attractive when the output is a class label and efficiency matters [7]. ModernBERT is a recent encoder designed specifically for efficient fine-tuning and inference [8].

The plan is therefore to compare a ModernBERT-based router with a simpler TF-IDF logistic-regression baseline. The baseline matters because the routing distinction may be partly lexical, since some question forms may simply correlate with graph need. ModernBERT tests whether pretrained semantic representations improve beyond those surface patterns.

2.4 Related Work

GraphRAG and LightRAG demonstrate that graph-enhanced retrieval can add structure beyond local vector similarity [2, 3]. They motivate the high-cost retrieval path, but neither answers the question of *when* graph-enhanced retrieval should be selected over a cheaper vector-based path.

Self-RAG and Adaptive-RAG show that retrieval behavior can be conditional rather than fixed [6, 4]. The present work follows that idea but applies it to a smaller action space: two retrieval modes in one Hybrid RAG framework rather than a general strategy selector.

RouterBench frames routing as a systems problem where the decision should account for both expected quality and cost [5], and recent encoder-only classification work supports using compact bidirectional models for efficient text classification [0, 7, 8]. What is missing from this literature is a focused study of LightRAG’s *Naive-versus-Mix* choice as a supervised, query-only routing problem. That is the gap the project addresses.

Chapter 3

Method

This chapter describes how the thesis will evaluate query-only retrieval-mode routing in a fixed Hybrid **RAG** setting. Section 3.1 presents the pipeline from 2WikiMultihopQA samples to router evaluation. Section 3.2 motivates the quantitative experimental approach. Section 3.3 describes the benchmark data, sampling, and target population. Section 3.4 defines the dual-mode LightRAG experiment and planned measurements. Section 3.5 discusses reliability and validity risks from judge labels, class imbalance, and benchmark scope. Section 3.6 presents the planned classifier and policy analysis. Section 3.7 defines how routing performance will be evaluated. Section 3.8 explains how the implementation will be documented.

3.1 Research Process

Figure 3.1 shows the end-to-end pipeline. It starts from a public multi-hop **QA** benchmark whose samples are processed and indexed in LightRAG while preserving the mapping between questions, answers, and supporting evidence [9, 3]. Each queryable sample is then executed with both *Naive* and *Mix* retrieval, the resulting answers are judged into routing classes, and the usable binary labels are used to train and evaluate query-only routers [10].

Step 1 Prepare 2WikiMultihopQA samples and preserve the mapping between questions, answers, context documents, and supporting facts.

Step 2 Index the relevant sample context documents in LightRAG.

Step 3 Execute every query in both retrieval modes and log answer text plus execution metadata.

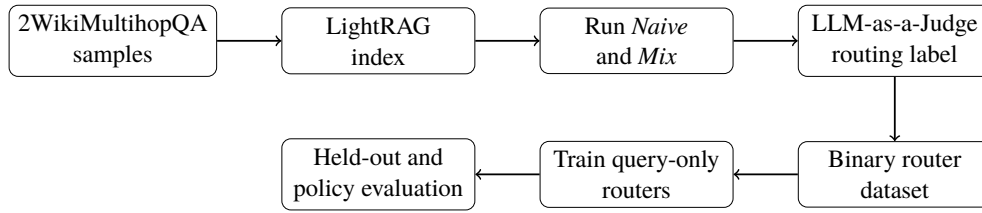


Figure 3.1: Planned research process for constructing and evaluating a learned retrieval-mode router.

Step 4 Construct routing labels with **LLM-as-a-Judge**.

Step 5 Train **TF-IDF** baseline and ModernBERT query routers.

Step 6 Evaluate the router both as a classifier and as a thresholded retrieval policy.

3.2 Research Paradigm

The study uses a quantitative experimental paradigm built around a controlled LightRAG setting where the same benchmark query is run with both *Naive* and *Mix* retrieval. Each query therefore produces two generated answers, a judged routing label, and a router prediction that can be compared against that label, making the routing problem directly observable.

The reasoning is mainly deductive. The starting assumption is that some queries contain enough information to indicate whether vector retrieval is likely sufficient or whether graph-enhanced retrieval is needed. If the router predicts the judged routing labels reliably and creates useful thresholded policies, the results support the assumption. If it performs close to trivial baselines, the results suggest that the retrieval-mode distinction is not predictable enough from query text alone in this setting.

The research question is about measurable system behavior, not about interviews, observations, or organizational practice. Ericsson documentation motivates why the problem matters industrially, but the core evidence comes from reproducible benchmark experiments, prediction metrics, and cost-performance comparisons.

3.3 Data Collection

The primary dataset is 2WikiMultihopQA [9], a public benchmark containing questions, answers, context documents, and supporting evidence. It was designed for multi-hop QA, where answering depends on combining information across reasoning steps, the kind of scenario where graph-enhanced retrieval is expected to matter most. Related benchmarks such as HotpotQA and MuSiQue share this motivation [13, 14]. No new human-subject data is collected, since the task is to transform the existing benchmark into a form suitable for retrieval-mode routing.

3.3.1 Sampling

The pipeline processes the benchmark in original sample order, converting each sample into a structured record containing the question, answer, question type, context documents, supporting facts, and evidence metadata. The LightRAG indexer then inserts context documents while tracking which samples are queryable. Preserving this mapping matters because each query result must later be connected back to its benchmark answer and routing label.

A held-out split of judged binary examples is used for router evaluation, stratified by routing label so that both `naive_enough` and `mix_required` examples appear in train, validation, and test partitions.

3.3.2 Sample Size

The binary routing dataset contains 1286 usable judged examples after excluding cases where neither retrieval mode was sufficient. Of these examples, 959 are labeled `naive_enough` and 327 are labeled `mix_required`. The dataset is split into 900 training examples, 193 validation examples, and 193 test examples.

The class distribution is imbalanced, with more examples where *Naive* retrieval is sufficient than examples where *Mix* is required. The evaluation will therefore report class-sensitive metrics such as AUPRC, precision, recall, F1, and balanced accuracy rather than relying only on overall accuracy [0].

3.3.3 Target Population

The immediate target population is 2WikiMultihopQA-style multi-hop questions over a controlled Wikipedia-derived corpus. The broader population

of interest is industrial **RAG** questions where some queries can be answered by cheap vector retrieval and others require structured retrieval. Whether results on 2Wiki transfer to Ericsson documentation or to other Hybrid **RAG** frameworks is an open question, and the conclusions will be scoped accordingly.

3.4 Experimental design and Planned Measurements

The experiment fixes the retrieval framework and varies only the routing decision. For each query q_i , both retrieval modes are executed.

$$r_i^N = R_N(q_i), \quad r_i^M = R_M(q_i), \quad (3.1)$$

where R_N denotes LightRAG *Naive*, R_M denotes LightRAG *Mix*, and r_i^N , r_i^M are the generated answers. A judge then assigns a class.

$$y_i = \begin{cases} 0, & r_i^N \text{ is sufficient,} \\ 1, & r_i^N \text{ is insufficient and } r_i^M \text{ is sufficient.} \end{cases} \quad (3.2)$$

A retrieval output is treated as sufficient when the judged answer matches the benchmark answer for the query. A query is labeled `naive_enough` when *Naive* is sufficient, `mix_required` when *Naive* is insufficient but *Mix* is sufficient, and `none_sufficient` when neither mode answers correctly. Examples where neither retrieval mode is sufficient are retained for analysis but excluded from the main binary router training set. This makes the binary label correspond to the minimum viable retrieval mode rather than general answer correctness, while still treating **LLM-as-a-Judge** as an imperfect evaluator [10].

3.4.1 Test environment and model

The test environment is a Python-based pipeline covering dataset preparation, LightRAG indexing, dual-mode querying, judge-based labeling, router training, and offline evaluation. LightRAG provides the **RAG** framework, with *Naive* as the low-cost vector path and *Mix* as the graph-enhanced hybrid path [3].

The router receives only the query text. It predicts a score

$$s_\phi(q_i) = P_\phi(y_i = 1 \mid q_i), \quad (3.3)$$

where $y_i = 1$ means that *Mix* is required. The primary learned model is a ModernBERT-based classifier with a lightweight classification head [8]. The query is tokenized and encoded by ModernBERT into a sequence representation h_i . The classification head maps this representation to two logits,

$$z_i = Wh_i + b, \quad (3.4)$$

where $z_{i,0}$ represents *Naive sufficient* and $z_{i,1}$ represents *Mix required*. The router score in Equation 3.3 is the softmax probability of the second class.

$$s_\phi(q_i) = \frac{\exp(z_{i,1})}{\exp(z_{i,0}) + \exp(z_{i,1})}. \quad (3.5)$$

The model is fine-tuned by minimizing cross-entropy loss on the judged routing labels,

$$\mathcal{L}(\phi) = -\frac{1}{n} \sum_{i=1}^n \log P_\phi(y_i \mid q_i). \quad (3.6)$$

Training uses backpropagation and the AdamW optimizer [0]. Since the binary classes are imbalanced, class weights can be used in the loss so that the minority `mix_required` class is not ignored during fine-tuning. The actual routing decision is not made by the classification head alone, but by applying a threshold to $s_\phi(q_i)$ during the policy evaluation described in Section 3.6.1. A **TF-IDF** logistic-regression model is used as a classical baseline.

3.4.2 Hardware/Software to be used

The implementation uses Python, LightRAG, PyTorch, Hugging Face Transformers, and scikit-learn. Gemini models handle LightRAG language-model calls, and `gemini-3-flash-preview` serves as the judge in the current setup. **GPU** access is useful for transformer fine-tuning, but the pipeline is designed to run incrementally so that data generation, baseline training, and offline evaluation do not have to happen in a single uninterrupted run.

Both retrieval-mode outputs and execution times are logged. Execution time is the practical cost proxy because it is available for both modes throughout the pipeline; `token usage` is treated as supplementary metadata.

A practical risk is that **API** cost and throughput can limit how quickly retrieval outputs and judge labels are produced, so the pipeline supports incremental execution and completed outputs are reused when a run is interrupted or extended.

3.5 Assessing reliability and validity of the data collected

3.5.1 Validity of method

The main construct validity risk is that judge labels may not perfectly represent true retrieval need. Prior work on **LLM-as-a-Judge** shows that language models can be useful evaluators, but also that their judgments need to be treated critically [10]. Constrained labels, grounding the judgment in the benchmark answer, and excluding examples where neither mode succeeds mitigate this, but the labels remain imperfect supervision.

An internal validity strength is that both retrieval modes run within the same framework over the same indexed corpus, which reduces confounding from comparing different **RAG** systems. The corresponding risk is that observed differences may partly reflect implementation details of LightRAG’s modes rather than graph retrieval in general.

A subtler validity risk comes from the stochastic nature of LightRAG graph construction and **LLM**-based generation. Entity extraction, relation extraction, summarization, and answer generation can all vary between runs even when the input documents are identical. If the graph were rebuilt several times, differences in router labels might partly reflect graph-construction variance rather than stable differences between *Naive* and *Mix*. To control for this, the experiment treats the constructed index and logged retrieval outputs as a fixed part of the experimental setup.

3.5.2 Reliability of method

Reliability requires that repeated runs lead to the same routing dataset, the same train/validation/test split, and similar router performance. Deterministic preprocessing, stable sample identifiers, fixed data splits, explicit random seeds, and saved prediction files all support this.

The main stochastic components are graph construction, answer generation, judge labeling, and neural router training. Separating data generation

from router evaluation reduces their impact. Once answers and judge labels have been produced, the router trains and evaluates on saved examples rather than on live **API** calls. Fixed splits and random seeds make repeated training runs directly comparable.

Stronger alternatives exist. Human expert annotation could provide more reliable routing labels, repeated judging with multiple models could estimate label uncertainty more directly, and rebuilding the LightRAG graph several times could measure graph-construction variance. These are outside the core scope, but constrained judge labels, exclusion of unclear examples, and held-out evaluation partly compensate.

The key reliability check is whether the router does more than exploit the majority class. If it only improves overall accuracy without improving **AUPRC**, balanced accuracy, and F1 for the *Mix required* class, the result is not reliable evidence for useful routing.

3.5.3 Data validity

Using a public benchmark strengthens data validity because questions and gold answers are externally defined rather than constructed to fit the router. That said, 2WikiMultihopQA is not Ericsson documentation, and the conclusions must be scoped to the benchmark setting unless an Ericsson extension is later added.

3.5.4 Reliability of data

Data reliability depends on preserving the link between each query, answer, context, retrieval outputs, judge label, and model configuration. The data pipeline is designed around stable sample identifiers to avoid mixing results across samples. Query and judge outputs are written incrementally so interrupted runs can resume without overwriting previous results.

3.6 Planned Data Analysis

3.6.1 Data Analysis Technique

The primary analysis evaluates **held-out** label prediction. Before choosing a threshold, **AUPRC** is used to measure how well the router ranks *Mix required* queries above *Naive sufficient* queries under class imbalance [0].

For thresholded evaluation, the router score $s_\phi(q_i)$ is converted into a class prediction with threshold τ .

$$\pi_\tau(q) = \begin{cases} M, & s_\phi(q) \geq \tau, \\ N, & s_\phi(q) < \tau. \end{cases} \quad (3.7)$$

Here M denotes *Mix* and N denotes *Naive*. The thresholded analysis reports accuracy, balanced accuracy, precision, recall, and F1 for the *Mix required* class.

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \quad (3.8)$$

The secondary analysis treats the router as an offline routing policy. For each threshold, the policy selects the already computed *Naive* or *Mix* result for each held-out query. The average routed time is defined as follows.

$$C(\pi_\tau) = \frac{1}{m} \sum_{i=1}^m [\mathbf{1}\{\pi_\tau(q_i) = N\}t_i^N + \mathbf{1}\{\pi_\tau(q_i) = M\}t_i^M], \quad (3.9)$$

where t_i^N and t_i^M are the measured execution times for *Naive* and *Mix*. This analysis makes it possible to compare the learned router with static policies such as always using *Naive* and always using *Mix*.

3.6.2 Software Tools

The software environment described in Section 3.4.2 supports all data analysis. The classifier evaluation compares the ModernBERT router with a TF-IDF logistic-regression baseline and a majority-class baseline, reporting both threshold-independent metrics (AUPRC) and thresholded metrics (precision, recall, F1, balanced accuracy). Offline policy analysis combines saved router predictions with the measured execution times for both retrieval modes.

All dataset splits, model predictions, metric calculations, and policy comparisons are saved so that reported tables and figures can be regenerated from the same inputs.

3.7 Evaluation framework

Evaluation has two sides. Classifier evaluation asks whether the ModernBERT router predicts held-out judged labels better than a **TF-IDF** logistic-regression baseline and a majority-class baseline, using **AUPRC** for ranking quality under class imbalance and thresholded metrics for concrete routing decisions. Routing-policy evaluation asks whether a thresholded router creates useful operating points between always-*Naive* and always-*Mix*. Both perspectives matter: a router with good classification metrics may still be unattractive if it routes too many examples to the expensive mode, or misses too many examples where *Mix* is required.

Threshold selection uses validation data only. Test data is reserved for final reporting of the selected threshold and threshold curves, reducing the risk of overfitting to the test set.

3.8 System documentation

The primary documentation is a maintained project repository with a clear README covering the purpose of the prototype, the required environment, the main pipeline steps, reproduction instructions, and output organization. For a software-based thesis this is the most useful form because it stays close to the code and can be updated as the implementation changes.

The report itself documents the system architecture, the data flow from benchmark sample to routing label, the evaluation procedure, and the main configuration choices. Detailed operational material can be placed in an appendix if needed.

The guiding standard is traceability. Any reported result should be traceable back to the dataset split, retrieval outputs, judge labels, router predictions, and evaluation procedure that produced it. Since the prototype is an experimental research system rather than a deployed service, this matters more than a separate operations manual.

Chapter 4

What you did

Choose your own chapter title to describe this

[Vad gjorde du? Hur gick det till? – Välj lämplig rubrik
("Genomförande", "Konstruktion", "Utveckling" eller annat)]

What have you done? How did you do it? What design decisions did you make? How did what you did help you to meet your goals?

Vad du har gjort? Hur gjorde du det? Vilka designval gjorde du?
Hur kom det du hjälpte dig att uppnå dina mål?

4.1 Hardware/Software design .../Model/Simulation model & parameters/...

Hårdvara / Mjukvarudesign ... / modell / Simuleringsmodell och parametrar / ...

Figure 4.1 shows a simple icon for a home page. The time to access this page when served will be quantified in a series of experiments. The configurations that have been tested in the test bed are listed in Table 4.1. In 7.0 % of cases, there was an error indicating xxxxx.

Figur 4.1 visar en enkel ikon för en hemsida. Tiden för att få tillgång till den här sidan när den laddas kommer att kvantifieras i en serie experiment. De konfigurationer som har testats i provbänk listas i tabell 4.1.

Vad du har gjort? Hur gjorde du det? Vilka designval gjorde du?

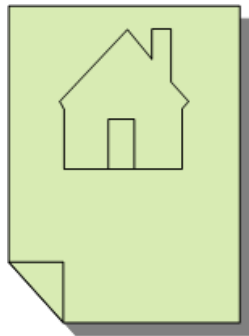


Figure 4.1: Homepage icon

Table 4.1: Configurations tested

Configuration	Description
1	Simple test with one server
2	Simple test with one server

Testade konfigurationer

4.2 Implementation .../Modeling/Simulation/...

Implementering ... / modellering / simulering / ...

Two commonly used simulators are:

- Mininet**

This simulator uses traffic control (tc) to simulate network devices connected by links with specific bandwidth, packet loss rates, qdisc methods, etc.
- ns-2 or ns-3 simulator**

These simulators are very useful for simulating wireless communication links between moving devices. You can specify the mobility patterns of the nodes.

4.2.1 Some examples of coding

This section is simply to show some example of how you can include code in your thesis - this is not a section you would have in your thesis.

Det här avsnittet är helt enkelt för att visa ett exempel på hur du kan inkludera kod i ditt examensarbete - det här är inte ett avsnitt du skulle ha i ditt examensarbete.

Listing 4.1 shows an example of a simple program written in C code.

Listing 4.1: Hello world in C code

```
int main() {
    printf("hello ,\world");
    return 0;
}
```

This template uses the package `lstlistings` for many different listings. Alternatively, one could use the `minited` package together with the `listings` environment, see, for example, [Code Highlighting with minted](#).

In contrast, Listing 4.2 is an example of code in Python to get a list of all of the programs at KTH. Note that on 2025-06-01 the KOPPS API will no longer work.

Note that the change to the iso date format was done in a group so that afterwards the date style returns to what it was, resulting in June 1, 2025.

Listing 4.2: Using a python program to access the KTH API to get all of the programs at KTH

```
KOPPSbaseUrl = 'https://www.kth.se '
```

```
def v1_get_programmes():
    global Verbose_Flag
    #
    # Use the KOPPS API to get the data
    # note that this returns XML
    url = "{0}/api/kopps/v1/programme".format(KOPPSbaseUrl)
    if Verbose_Flag:
        print("url:\n" + url)
    #
```

```

r = requests.get(url)
if Verbose_Flag:
    print("result_of_getting_v1_programme:{}".format(r.text))
#
if r.status_code == requests.codes.ok:
    return r.text          # simply return the XML
#
return None

```

In Listing 4.3, line 5 shows the use of the `ExcelWriter` function. Line 13 writes a panda dataframe (named `comp`) to the spreadsheet, while line 19 closes the open spreadsheet.

Listing 4.3 Example of using minted with python code

```

1 import pandas as pd
2 def make_spreadsheet_of_differences(pds):
3     global school
4     publishers=set()
5     writer =
        ↪ pd.ExcelWriter(f'/tmp/{school}_compare_duplicates.xlsx',
        ↪ engine='xlsxwriter')
6     print("starting")
7     for idx, p in enumerate(pds):
8         p0=list(p)[0]
9         p1=list(p)[1]
10        print(f"\nfor {p0} and {p1}")
11        comp, publisher1,
            ↪ publisher2=compare_two_records_silent(p0,
            ↪ p1)
12        sheet_name=f'{p0}'.split(':')[1]
13        print(f'Wrote {sheet_name=}')
14        comp.to_excel(writer,
            ↪ sheet_name=sheet_name)
15        publishers.add(publisher1)
16        publishers.add(publisher2)
17
18        # Close the Pandas Excel writer and output the
        ↪ Excel file.
19    writer.close()
20    print(f'{publishers=}')
21    return publishers

```

4.2.2 Some examples of figures in tikz

This section is simply to show some example of how you can draw your own figures for in your thesis - this is not a section you would have in your thesis.

Det här avsnittet är helt enkelt för att visa ett exempel på hur du kan rita dina egna figurer i ditt examensarbete – det här är inte ett avsnitt du skulle ha i ditt examensarbete.

These figures are just some examples to show that you can draw your own figures for your thesis. This has two advantages: (i) you do not have to worry about copyrights – as these are your own figures and (ii) the text is now readable and not simply a picture of text – so screen readers can read the figure’s contents to someone who is listening to the contents of your thesis.

4.2.2.1 Azure’s Form Recognizer

Figure 4.2 shows the processing of key-value extraction from a PDF document using Azure’s Form Recognizer.

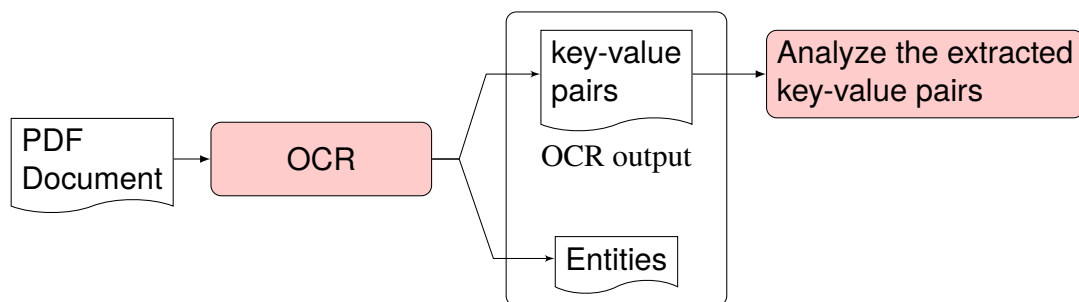


Figure 4.2: The processing of key-value extraction from a PDF document using Azure’s Form Recognizer

4.2.2.2 Hyper-V with Containers

Figure 4.3 shows how Hyper-V deals with containers.

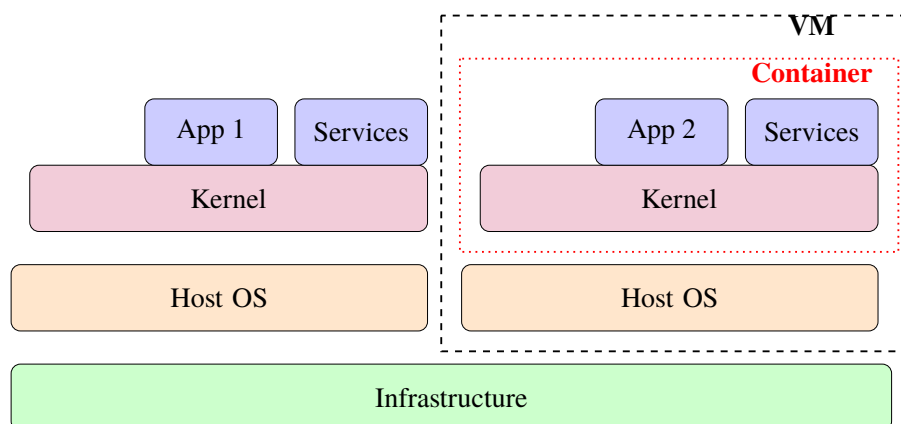


Figure 4.3: Hyper-V with containers

4.2.2.3 VM versus Containers

Figure 4.4 shows a comparison of virtual machines (VMs) versus containers.

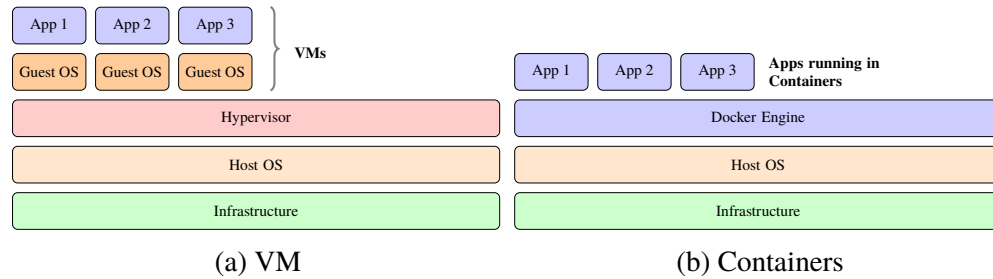


Figure 4.4: Virtual machines (VMs) versus Containers

Chapter 5

Results and Analysis

svensk: Resultat och Analys

Sometimes this is split into two chapters.

Keep in mind: How you are going to evaluate what you have done?

What are your metrics?

Analysis of your data and proposed solution

Does this meet the goals which you had when you started?

In this chapter, we present the results and discuss them.

I detta kapitel presenterar vi resultaten och diskutera dem.

Ibland delas detta upp i två kapitel.

Hur du ska utvärdera vad du har gjort? Vad är din statistik?

Analys av data och föreslagen lösning

Innebär detta att uppfyllelse av de mål som du hade när du började?

5.1 Major results

Huvudsakliga resultat

Some statistics of the delay measurements are shown in Table 5.1. The delay has been computed from the time the GET request is received until the response is sent.

Lite statistik av fördröjningsmätningarna visas i Tabell 5.1. Förseningen har beräknats från den tidpunkt då begäran GET tas emot fram till svaret skickas.

Table 5.1: Delay measurement statistics

Configuration	Average delay (ns)	Median delay (ns)
1	467.35	450.10
2	1 687.5	901.23

Table 5.2 shows the measurement of round trip times from four hosts to and from a server.

Table 5.2: Result for the ping measurements of RTT for 4 hosts

Host	host to server RTT in ms			
	min	avg	max	mdev
h1	5.625	5.625	5.625	0.0
h2	2.909	2.909	1.909	0.0
h3	5.007	5.007	5.007	0.0
h4	2.308	2.308	2.308	0.0

Fördröj mätstatistik

Konfiguration | Genomsnittlig fördröjning (ns) | Median fördröjning (ns)

Figure 5.1 shows an example of the performance as measured in the experiments.

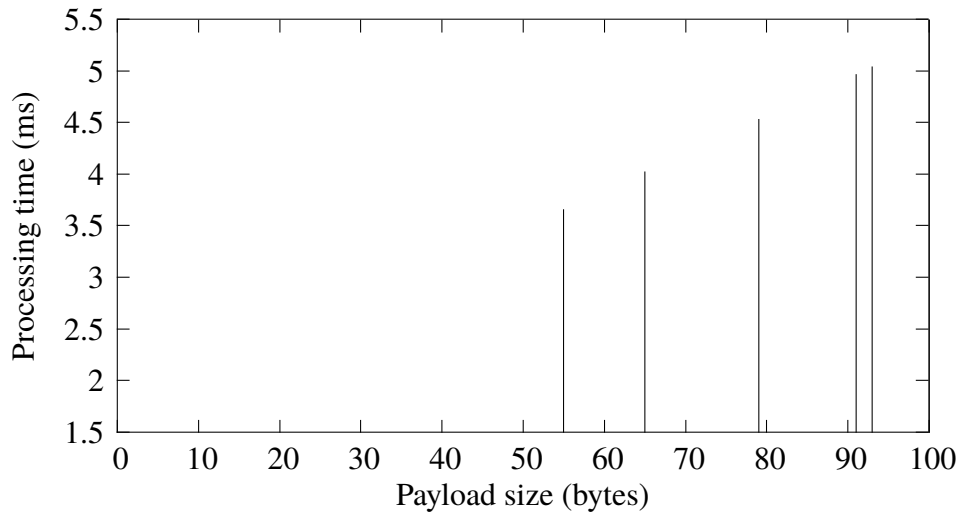


Figure 5.1: Processing time vs. payload length

Given these measurements, we can calculate our processing bit rate as the inverse of the time it takes to process an additional byte divided by 8 bits per byte:

$$\text{bit rate} = \frac{1}{\frac{\text{time}_{\text{byte}}}{8}} = 20.03 \text{ kb/s}$$

Table 5.3 shows another table in which some values have been set in bold (using \B) to emphasize them. Note how the S formatting has been modified so that it considers the weight of the characters and this is able to decimal align even these hold-faced numbers with the numbers in the column above them.

Table 5.3: Median values of sandwich attributes

Attribute	sites	
	A	B
price (in SEK)	36.5	71.3
protean (g)	97.2	100.0
salt (mg)	9.7	9.3
Average customer rating in %	82.2	89.9

Figure 5.2 shows a stacked bar chart using pgfplots. It illustrates how easy it is to take a set of data and make a stacked bar plot. One of the features is the shifted values – this is very useful when the bar itself is too small to put the value into.

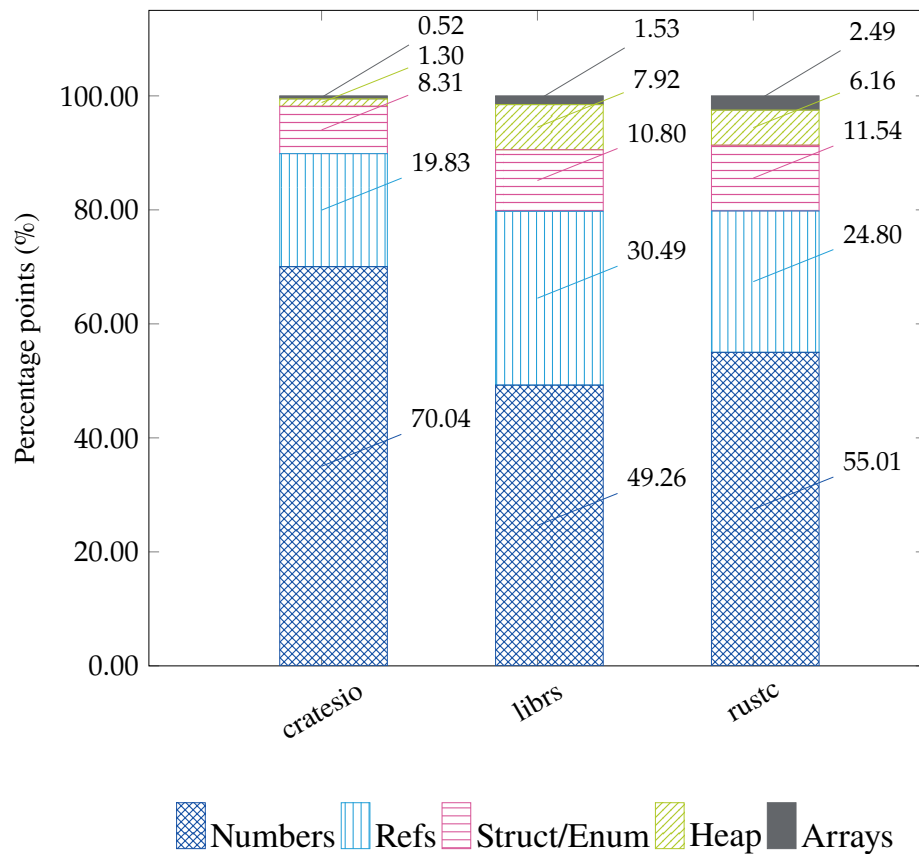


Figure 5.2: Rust types distribution for the compiler, crates.io, and lib.rs. (percentage) - appears here with the permission of the author - see the thesis at <https://urn.kb.se/resolve?urn=urn%3Anbn%3Ase%3Akt%3Adiva-332124>

5.2 Reliability Analysis

Analys av tillförlitlighet
Tillförlitlighet i metod och data

5.3 Validity Analysis

Analys av validitet
Validitet i metod och data

Chapter 6

Discussion

Diskussion
Förbättringsförslag?

This can be a separate chapter or a section in the previous chapter.

Chapter 7

Conclusions and Future work

Slutsats och framtida arbete

Add text to introduce the subsections of this chapter.

7.1 Conclusions

Slutsatser

Describe the conclusions (reflect on the whole introduction given in Chapter 1).

Discuss the positive effects and the drawbacks.

Describe the evaluation of the results of the degree project.

Did you meet your goals?

What insights have you gained?

What suggestions can you give to others working in this area?

If you had it to do again, what would you have done differently?

Uppfyllde du dina mål?

Vilka insikter har du fått?

Vilka förslag kan du ge till andra som arbetar inom detta område? Om du skulle göra detta igen, vad skulle du ha gjort annorlunda?

7.2 Limitations

Begränsande faktorer

Vad gjorde du som begränsade dina ansträngningar? Vilka är begränsningarna i dina resultat?

What did you find that limited your efforts? What are the limitations of your results?

7.3 Future work

Vad du har kvar ogjort?

Vad är nästa självklara saker som ska göras?

Vad tips kan du ge till nästa person som kommer att följa upp på ditt arbete?

Describe valid future work that you or someone else could or should do. Consider: What you have left undone? What are the next obvious things to be done? What hints can you give to the next person who is going to follow up on your work?

Due to the breadth of the problem, only some of the initial goals have been met. In these section we will focus on some of the remaining issues that should be addressed in future work. ...

7.3.1 What has been left undone?

The prototype does not address the third requirement, *i.e.*, a yearly unavailability of less than 3 minutes; this remains an open problem. ...

7.3.1.1 Cost analysis

Example of a missing component

The current prototype works, but the performance from a cost perspective makes this an impractical solution. Future work must reduce the cost of this solution; to do so, a cost analysis needs to first be done. ...

7.3.1.2 Security

Example of a missing component

A future research effort is needed to address the security holes that results from using a self-signed certificate. Page filling text mass. Page filling text mass. ...

7.3.2 Next obvious things to be done

In particular, the author of this thesis wishes to point out xxxxxx remains as a problem to be solved. Solving this problem is the next thing that should be done. ...

7.4 Reflections

Reflektioner

Vilka är de relevanta ekonomiska, sociala, miljömässiga och etiska aspekter av ditt arbete?

What are the relevant economic, social, environmental, and ethical aspects of your work?

One of the most important results is the reduction in the amount of energy required to process each packet while at the same time reducing the time required to process each packet.

The thesis contributes to the **United Nations (UN) Sustainable Development Goals (SDGs)** numbers 1 and 9 by xxxx.

In the references, let Zotero or other tool fill this in for you. I suggest an extended version of the IEEE style, to include URLs, DOIs, ISBNs, etc., to make it easier for your reader to find them. This will make life easier for your opponents and examiner.

IEEE Editorial Style Manual: https://www.ieee.org/content/dam/ieee-org/ieee/web/org/conferences/style_references_manual.pdf

Låt Zotero eller annat verktyg fylla i det här för dig. Jag föreslår en utökad version av IEEE stil - att inkludera webbadresser, DOI, ISBN osv. - för att göra det lättare för läsaren att hitta dem. Detta kommer att göra livet lättare för dina opponenter och examinerator.

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If you do not have an appendix, do not include the `\cleardoublepage` command below; otherwise, the last page number in the metadata will be one too large.

Appendix A

Supporting materials

Here is a place to add supporting material that can help others build upon your work. You can include files as attachments to the PDF file or indirectly via URLs. Alternatively, consider adding supporting material uploaded as separate files in DiVA.

The BibTeX references used in this thesis are attached. 

Some source code relevant to this project can be found at <https://github.com/ggmaquirejr/E-learning> and <https://github.com/ggmaquirejr/Canvas-tools>.

Your reader can access the attached (embedded) files using a PDF tool such as Adobe Acrobat Reader using the paperclip icon in the left menu, as shown in Figure A.1 or by right-clicking on the push-pin icon in the PDF file and then using the menu to save the embedded file as shown in Figure A.2.

An argument for including supporting material in the PDF file is that it will be available to anyone who has a copy of the PDF file. As a result, they do not have to look elsewhere for this material. This comes at the cost of a larger PDF file. However, the embedded files are encoded into a compressed stream within the PDF file; thus, reducing the number of additional bytes. For example, the references.bib file that was used in this example is 10 617 B in size but only occupies 4 261 B in the PDF file.

DiVA is limited to ≈ 1 GB for each supporting file. If you have very large amounts of supporting material, you will probably want to use one of the data repositories. For additional help with this, contact KTH Library via researchdata@kth.se.

As of Spring 2024, there are plans to migrate this supporting data from DiVA to a research data repository.

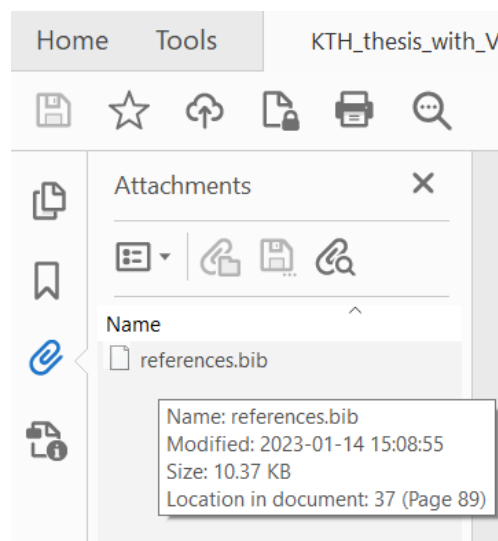


Figure A.1: Adobe Acrobat Reader using the paperclip icon for the attached references.bib file

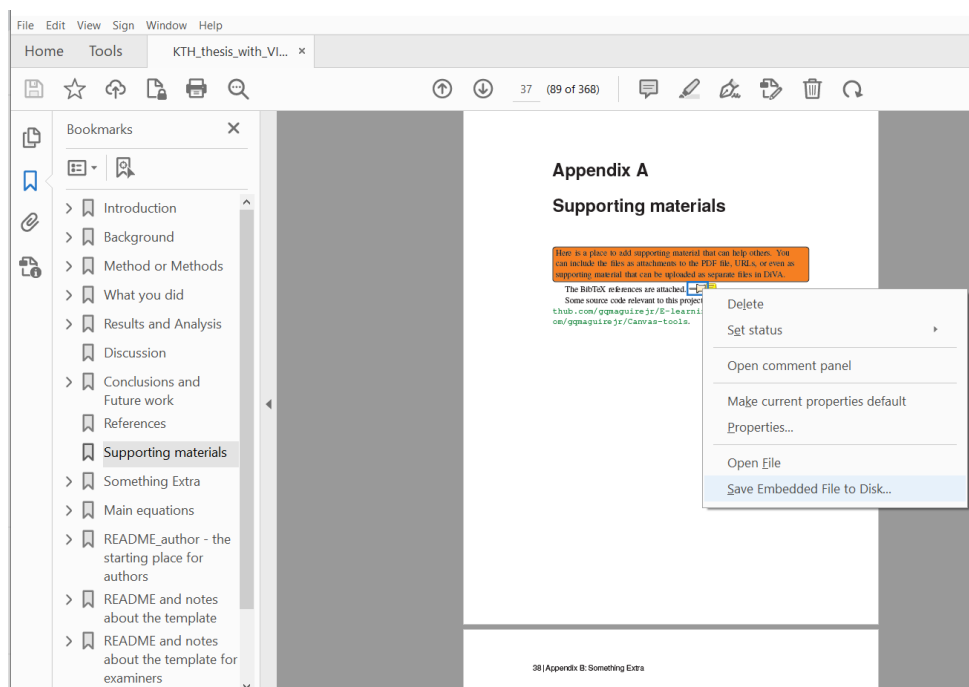


Figure A.2: Adobe Acrobat Reader after right-clicking on the push-pin icon for the attached references.bib file

Appendix B

Something Extra

svensk: Extra Material som Bilaga

B.1 Just for testing KTH colors

You have selected to optimize for print output

- Primary color

- kth-blue 

- kth-blue80 

- Secondary colors

- kth-lightblue 

- kth-lightred 

- kth-lightred80 

- kth-lightgreen 

- kth-coolgray 

- kth-coolgray80 

black 

Appendix C

README_author - the starting place for authors

This chapter describes the thesis template that I (Gerald Q. Maguire Jr.) have developed for use at KTH Royal Institute of Technology (KTH). It is important to note that the template is **not prescriptive**, as not every thesis will have all of the parts that the template shows. However, if there is something that you decide to leave out, you should make a conscious decision to do so and you should consider the impact this may have on your thesis being approved by the examiner.

Fundamental to the design of the template are several key factors:

- Helping students be successful in their degree project,
- Helping students produce a high-quality thesis, and
- Supporting all of the (relevant) phases of the degree project process.

This document is a work in progress.

C.1 Advice for Author or Authors

One of the hardest problems an author faces is getting started writing, *i.e.*, the blank sheet of paper – empty file barrier. The template provides a non-blank starting point; hence, avoiding the blank paper barrier. Additionally, the template provides some initial structure, basically, an Introduction, Methods, Results, and Discussion (IMRAD) structure, so that there are hints of where to place material. Moreover, there are places (and notes) about material that the

student should consider adding; for example, the “required reflections” section in the final chapter.

The template (located in the file `examplethesis.tex`) also provides some examples of commonly occurring types of content, so that one can easily find examples of how to include a figure, table, code listing, *etc.* These examples are not meant to be exhaustive and quite often the student will probably need to learn new $\text{\LaTeX}$ commands in the course of writing their thesis.

As an author, the first step is to configure the $\text{\LaTeX}$ engine that you will use to process the files - see Section C.2. The second step will be to configure the template - see Section C.3. The third step will be to make sure that the information about you, your supervisor(s), and the examiner are correct in the file `custom_configuration.tex` - this information uses the macros described in Section C.4. Now that you have a lot of the administrative details taken care of it is time to start to write - see Section C.6.

Note that if you are using Overleaf:

Make your own copy of the template If you have opened the template from a URL, in the upper left-hand corner, click on **Menu**. Then select **Copy Project** - this will give you your own private copy.

Use a helpful project name I suggest you include your name in the project name so that when you share it with your supervisor(s) and examiner, they will know it is your project.

Invite your supervisor(s) and examiner to your project You can invite your supervisor(s) and examiner to your project and they can directly comment on and correct your drafts.

Log in to Overleaf with your KTH account If you log in to Overleaf with your KTH account, you get a version of Overleaf that lets you turn on “Track changes” which is very useful (particularly if you have invited your supervisor(s) and examiner to join your project). It also gives you a bit more of a time budget to compile (which can be useful if you have a lot of Tikz figures or other things that take a lot of time for the $\text{\LaTeX}$ engine to render).

If you have more detailed questions about the template itself - see Chapter D.

C.2 Author configuration of the $\text{\LaTeX}$ engine

The template should work with $\text{\pdfLaTeX}$, $\text{\XeLaTeX}$, and $\text{\LuaLaTeX}$. If you are using Overleaf, I strongly recommend using $\text{\XeLaTeX}$ — as this will get the `Arial` fonts correct for the KTH cover. If you are running the compiler on your local machine and you use $\text{\XeLaTeX}$ **and** you have `Arial` as a system font, then it will be able to use it. Similarly, for $\text{\LuaLaTeX}$. For $\text{\pdfLaTeX}$ I have used `\fontfamilyhelvet`, *i.e.*, Helvetica, as it is a sans serif font.

One student reported problems with `FONTSPEC` not loading the fonts properly when running locally with macOS 12.4, TeXLive 2022, LaTeX Workshop on VS Code, and $\text{\XeLaTeX}$ - the solution is described at <https://tug.org/TUGboat/tb39-2/tb122robertson-fontspec.pdf>.

If you are using Overleaf, it is easy to select the compiler (*i.e.*, $\text{\TeX}$ engine) by using the drop-down menu, as shown in Figure C.1.

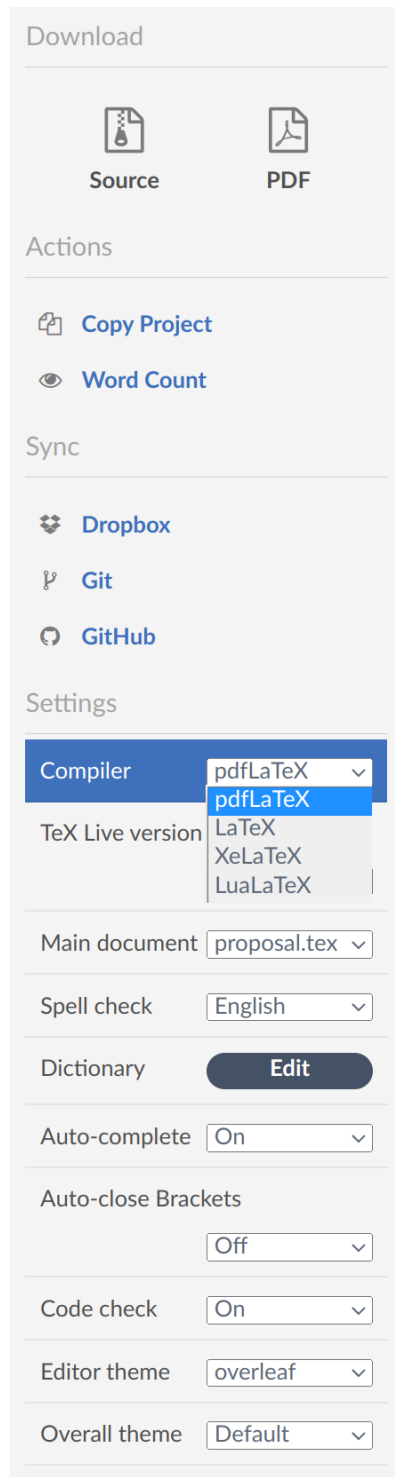


Figure C.1: Selecting a compiler (*i.e.*, TeX engine) in Overleaf

C.3 Author configuration of the template

The template is designed to handle a thesis written in English or Swedish. You can set the default language to ‘english’ or ‘swedish’ by passing an option to the documentclass. Note that the language option is written in all lowercase letters; for example, to set the document’s language to English:

```
\documentclass[english]{kththesis}
```

To set the document’s language to Swedish (uncomment the following line):

```
\documentclass[swedish]{kththesis}
```

The language option ‘swedish’ sets the conditional `\ifinswedish` to true. Among many other things, this conditional is used to configure the KTH cover and the title page to use the chosen language.

The two most common bibliographic engines are supported, *i.e.*, BibTeX and BibLaTeX. To set the language to English and use the bibliographic engine to BibTeX you would say:

```
\documentclass[english, bibtex]{kththesis}
```

To set the language to Swedish and use the bibliographic engine to BibLaTeX you would say:

```
\documentclass[swedish, biblatex]{kththesis}
```

The above illustrates that you can pass multiple options to the document class separated by commas. Also, note that the options were passed as all lowercase letters.

You can, of course, also modify the formatting of the citations and bibliography. See for example the following code snippet:

```
\ifbiblatex
% \usepackage[language=english, bibstyle=authoryear,
% ↪ citestyle=authoryear, maxbibnames=99]{biblatex}
% \usepackage[style=numeric, sorting=none, backend=biber
% ↪ ]{biblatex}
\usepackage[bibstyle=authoryear, citestyle=authoryear,
% ↪ maxbibnames=99, language=english]{biblatex}
% alternatively you might use another style, such as
% ↪ IEEE
% \usepackage[style=ieee]{biblatex}
\addbibresource{references.bib}
% \DeclareLanguageMapping{norsk}{norwegian}
```

```
\else
  % The line(s) below are for BibTeX
  \bibliographystyle{bibstyle/myIEEETran}
  %\bibliographystyle{apalike}
\fi
```

To optimize for digital output (this changes the color palette) add the option: `digitaloutput`. There are also options for A4 or G6 paper: `a4paper` or `g5paper` (respectively). There is an option for nomenclature, to produce and refer to equations. Finally, there are options for a 1st cycle thesis or 2nd cycle thesis: `bachelor` and `master` (respectively); however, these two options are **not** currently used.

One of the first things that the author(s) will want to do is add the working title and subtitle to the thesis. This is done using the `\title`, `\subtitle`, `\alttitle`, and `\altsubtitle` macros as shown below:

```
\title{This is the title in the language of the thesis}
\subtitle{A subtitle in the language of the thesis}

% give the alternative title - i.e., if the thesis is in
  ↪ English,
% then give a Swedish title
\alttitle{Detta är den svenska översättningen av titeln}
\altsubtitle{Detta är den svenska översättningen av
  ↪ undertiteln}
% alternative, if the thesis is in Swedish, then give an
  ↪ English title
%\alttitle{This is the English translation of the title}
%\altsubtitle{This is the English translation of the
  ↪ subtitle}
```

Setting these values once and then using them in many places reduces the work to change them while at the same time ensuring consistency.

The issue of what characters can be placed in a title (subtitle, keywords, and abstracts) and how to enter them is a bit complex, but you should be able to achieve much of what you want - at the cost of some effort. Ultimately, the characters that will be placed into DiVA and LADOK must be Unicode, encoded using Unicode Transformation Format (UTF), specifically UTF-8. While DiVA utilizes HTML for these fields, allowing you to include some markup, LADOK does not support any markup.

If you have titles or subtitles that cannot be converted to pure Unicode (such as subscripts with capital letters or other symbols), you should also use the optional commands `\titleInPlainText`, `\subtitleInPlainText`, `\alttitleInPlainText`, and `\altsubtitleInPlainText` in your `custom_configuration.tex` or `custom_configuration_plaintext.tex` files to provide versions of titles and subtitles that can be used with LADOK. Compiling `Generate_titles_in_plain_text.tex` with **Lua^AT_EX** will produce a file named `extra_plaintext.tex` containing plain text versions of the titles and subtitles (both main and alternate). The reason that you have to compile using Lua^AT_EX is that I have used Lua code and Lua^AT_EX to compute the plain text strings. After you have compiled the file, you can follow the workflow to generate, download, edit and merge, and upload the `custom_configuration_plaintext.tex` file to your Overleaf project.

Some additional configuration that the author(s) might do is to set the values of the macros related to the course cycle, course code, date of the thesis, number of credits, degree/exam name, subject area, and if the degree is done external to KTH to set the host information (see the file *custom_configuration.text*). Consider the snippet below for a student admitted to the “Bachelor’s Programme in Information and Communication Technology (TCOMK)” program and enrolled in the degree project course “IA150X Degree Project in Information and Communication Technology, First Cycle 15.0 credits” and working at a company “Företaget AB”:

```
\hostcompany{Företaget AB} % Remove this line if the
    ↳ project was not done at a host company

\date{\today}

\courseCycle{1}
\courseCode{IA150X}
\courseCredits{15.0}

\programcode{TCOMK}
\degreeName{Bachelors degree}
% Note that the subject area for a Bachelor's thesis (
    ↳ Kandidatexamen)
```

```
% should be either Technology or Architecture
% If the thesis is in Swedish, these would be: teknik |
    ↪ arkitektur
% -- Note the use of lower case for the Swedish subject
    ↪ area
\subjectArea{Technology}
```

Note that in the above macros you have to give the English or Swedish names in the arguments to \degreeName and \subjectArea - as shown below:

```
\degreeName{Kandidatexamen}
\subjectArea{teknik}
```

For a CDATE student enrolled in the course “DA231X Degree Project in Computer Science and Engineering, Second Cycle 30.0 credits”, the cycle, program, course code, degree, and subject area information would be:

```
\programcode{CDATE}
\courseCycle{2}
\courseCode{DA231X}
\courseCredits{30.0}
\degreeName{Degree of Master of Science in Engineering}
\subjectArea{Computer Science and Engineering}
```

The set of possible values for the English or Swedish names in the arguments to \degreeName are:

```
\degreeName{Higher Education Diploma}
\degreeName{Högskoleexamen}

\degreeName{Bachelors degree}
\degreeName{Kandidatexamen}

\degreeName{Master of Architecture}
\degreeName{Arkitektexamen}

\degreeName{Degree of Master of Science in Engineering}
\degreeName{Civilingenjörs}

\degreeName{Magister}
\degreeName{Magisterexamen}

\degreeName{Degree of Master of Science}
\degreeName{Masterexamen}
```

```

\degreeName{Master of Science in Engineering and Master
↪ of Arts in Education degree}
\degreeName{Civilingenjör och lärare examen}

\degreeName{Degree of Master of Science in Secondary
↪ Education}
\degreeName{Ämneslärarexamen}

\degreeName{Both} # Degree Project in the Field of
↪ Technology <teknikområde> and the Main Field of
↪ Study <huvudområde>
\degreeName{Same} # The case when the field of
↪ technology <teknikområde> and main field of study <
↪ huvudområde> are the same.

```

For the last two cases, the code compares the values of subjectArea and secondSubjectArea.

You can find a list of the program codes and school acronyms in the file: lib/schools_and_programs.ins.

There are a set of rules about what is to be displayed on the KTH cover. These can be found at <https://www.kth.se/social/group/sprakkommitte/page/omrade-for-examensarbete/>.

One of the reasons for many of the macros shown above and below is to collect the information that is needed to report the approved thesis in Digitala Vetenskapliga Arkivet (DiVA) and to report the title(s) and grade in Lokalt adb-baserat dokumentationssystem (LADOK).

National subject categories are a **required** field in the DiVA record. These categories follow a definition by SCB (nowadays known as Statistikmyndigheten or in English: Statistics Sweden) and HSV (Högskoleverket - nowadays known as Universitetskanslersämbetet (UK-ämbetet) and Universitets- och högskolerådet (UHR) or in English: Swedish Higher Education Authority and Swedish Council for Higher Education). While these codes refer to research areas, these codes are also used in KTH to indicate the area of the thesis. The guidance that I received from the Linköping University library was that one should try to use 5-digit codes when possible. Some examples of these codes are shown in Table C.1.

```

\nationalsubjectcategories{} comma-separated list of national
                             subject category codes - each a 3
                             or 5 digit code

```

An example for a thesis in Computer Science and Computer Systems:

```
\nationalsubjectcategories{10201, 10206}
```

You can find examples of the subjects and their codes at in Table C.1. The complete list can be found AT <https://www.scb.se/contentassets/3a12f556522d4bdc887c4838a37c7ec7/standard-for-svensk-indelning--av-forskningsamnen-2011-uppdaterad-aug-2016.pdf>

and

<https://www.scb.se/contentassets/10054f2ef27c437884e8cde0d38b9cc4/oversattningsnyckel-forskningsamnen.pdf>

You can also find them in README_notes/National_Subject_Categories.tex. This file can be processed separately or included as an appendix by commenting out the begin and end comments around it near the end of the examplethesis.tex file. The full list is about 14 pages long.

Table C.1: Examples of some national subject categories and their codes

Code	Category (in Swedish)	Category (in English)
102	Data- och informationsvetenskap (Datateknik)	Computer and Information Sciences
10201	Datavetenskap (datalogi)	Computer Sciences
10202	Systemvetenskap, informationssystem och informatik (samhällsvetenskaplig inriktning under 50804)	Information Systems (Social aspects to be 50804)
10203	Bioinformatik (beräkningsbiologi) (tillämpningar under 10610)	Bioinformatics (Computational Biology) (applications to be 10610)
10204	Människa-datorinteraktion (interaktionsdesign) (Samhällsvetenskapliga aspekter under 50803)	Human Computer Interaction (Social aspects to be 50803)
10205	Programvaruteknik	Software Engineering
10206	Datorteknik	Computer Engineering
10207	Datorseende och robotik (autonoma system)	Computer Vision and Robotics (Autonomous Systems)
10208	Språkteknologi (språkvetenskaplig databehandling)	Language Technology (Computational Linguistics)
10209	Medieteknik	Media and Communication Technology
10299	Annan data- och informationsvetenskap	Other Computer and Information Science
202	Elektroteknik och elektronik	Electrical Engineering, Electronic Engineering, Information Engineering
20201	Robotteknik och automation	Robotics
20202	Reglerteknik	Control Engineering
20203	Kommunikationssystem	Communication Systems
20204	Telekommunikation	Telecommunications
20205	Signalbehandling	Signal Processing
20206	Datorsystem	Computer Systems
20207	Inbäddad systemteknik	Embedded Systems
20299	Annan elektroteknik och elektronik	Other Electrical Engineering, Electronic Engineering, Information Engineering

C.3.1 Setting the UN's Sustainable Development Goals (SDGs) information

The UN’s Sustainable Development Goals (SDGs) are a field in the DiVA record. Some examples of these SDGs are shown in Table C.2.

\SDGs{ }	comma-separated list of SDG codes - each a numerical value
-----------------	---

An example of a thesis supporting SDGs 8 and 9:

$$\backslash \text{SDGs} \{8, 9\}$$

Table C.2: UN's Sustainable Development Goals (SDGs)

SDG	Category (in English)	Category (in Swedish)
1	No poverty	Ingen fattigdom
2	Zero hunger	Ingen hunger
3	Good health and well-being	God hälsa och välbefinnande
4	Quality education	God utbildning för alla
5	Gender equality	Jämställdhet
6	Clean water and sanitation	Rent vatten och sanitet för alla
7	Affordable and clean energy	Hållbar energi för alla
8	Decent work and economic growth	Anständiga arbetsvillkor och ekonomisk tillväxt
9	Industry, innovation and infrastructure	Hållbar industri, innovationer och infrastruktur
10	Reduced inequalities	Minskad ojämlikhet
11	Sustainable cities and communities	Hållbara städer och samhällen
12	Responsible consumption and production	Hållbar konsumtion och produktion
13	Climate action	Bekämpa klimatförändringarna
14	Life below water	Hav och marina resurser
15	Life on land	Ekosystem och biologisk mångfald
16	Peace, justice and strong institutions	Fredliga och inkluderande samhällen
17	Partnerships for the goals	Genomförande och globalt partnerskap

C.4 Author macros

It is assumed that there can only be 1 or 2 authors. For many years now 2nd cycle theses are expected to only have one author.

For the author or first author, there are a number of macros defined to store information about the author, so that it can later be used in multiple places – for example, the KTH cover (produced with `\kthcover`), the title page (via `\titlepage`, the “For DiVA” section at the end of the thesis (via `\divainfo{pg:lastPageofPreface}{pg:lastPageofMainmatter}`), and possibly a JavaScript Object Notation (JSON) file named `fordiva.json` produced as a by product of the `\divainfo`. Note that the actual section name has DiVA set in all caps - which hopefully should not occur in the thesis! If the string DiVA set in all caps, does have to appear, then the section heading should be preceded by four euro signs and followed by four more euro signs (as is done in this document).

The author-related macros are:

<code>\authorsLastname{ }</code>	the last name of the author*
<code>\authorsFirstname{ }</code>	the first name of the author
<code>\email{ }</code>	the KTH e-mail address of the author
<code>\kthid{ }</code>	the author’s kthid, this generally starts with the string “u1” and is a unique identifier for every KTH user.
<code>\authorsSchool{ }</code>	the value is generally of the form: <code>\schoolAcronym{XXX}</code> . The currently supported school acronyms are: ABE, CBH, EECS, ITM, and SCI. These are defined in the file <code>schools_and_programs.ins</code> . There is also a fake school XXX for use in the template.

If the first author is not in Stockholm, Sweden when the acknowledgements are written, then add that information via the macros described below. This information will be used when generating the acknowledgements

*Note that the author’s name can include a suffix such as “, Jr.” or “ Jr.”, i.e., the suffix can be separated with a comma or not – as the author prefers to write their name.

signature. The acknowledgements signature is the text at the end of the acknowledgements and it gives the place where the author(s) is/are when writing the acknowledgements and also gives the date and name(s).

`\authorCity{A City}` specify the city

`\authorCountry{A Country}` specify the country

`\authorCityCountryDate{}` pass into this function the month and year for the acknowledgement. This can be a string such as January 2022 or it can be a L^AT_EX expression, such as `\MONTH\enspace\the\year`.

If there is a second author and the place, month, and year are **all** the same, then specify the month and year for only the **first** author:

`\authorCityCountryDate{\MONTH\enspace\the\year}`

If there is a second author and the place is different, then say:

`\authorCityCountryDate{}`

If there is a second author, the macros are:

`\secondAuthorsLastname{}` the last name of the 2nd author

`\secondAuthorsFirstname{}` the first name of the 2nd author

`\secondemail{}` the KTH e-mail address of the 2nd author

`\secondkthid{}` the 2nd author's kthid

`\secondAuthorsSchool{}` the school of the 2nd author

If the second author is not in the same place as the first author, then add the relevant information using the macros below. This information will be used when generating the acknowledgements signature.

`\secondAuthorCity{A City}` specify the city

`\secondAuthorCountry{A Country}` specify the country

`\secondAuthorCityCountryDate{\MONTH\enspace\the\year}`
pass into this function the month and year for the acknowledgement

If the second author is the same place as the first author, then comment out or delete the `\secondAuthorCityCountryDate{ }` as shown below:

```
%\secondAuthorCityCountryDate{ }
```

C.5 Starting to write - for the impatient

For those who are impatient and rapidly want to start writing, I suggest you start by configuring the `custom_configuration.tex` file (see Section C.3), your working title, and abstract (Section C.6.1). After this, a quick way to start writing the text in your document is to go to the table of contents in Overleaf and click on a chapter or section - this will utilize a hyperlink to go to that part of the PDF file. Next, click on the left-going arrow near the top of the border between the LaTeX on the left and the PDF on the right; this will take you (close) to the correct place in the source file where you can start to modify the content and write.

C.6 Starting to write

As you write you will notice "todo" notes in the template. They follow the following conventions:

```
\generalExpl{Comments/directions/... in English}
\sweExpl{Text på svenska}
\engExpl{English descriptions about formatting}
\sweExpl{warnings}
```

If you do not want to see these notes, you can, of course, redefine the above macros to output nothing. If you do not want to see any notes, then add the option **final** to the `\documentclass` arguments near the top of the file `examplethesis.tex`.

C.6.1 Working abstract

I generally recommend that every student start by writing a working abstract, this will help you keep your focus. To find where you can start to enter your abstract, look in the `examplethesis.tex` file for the line:

```
\generalExpl{Enter your abstract here!}
```

There is lots of information already in the template to help you with entering text, equations, *etc.*, in your abstract. **NB** Abstracts are supposed to

stand by themselves, this means no footnotes, no cross-references, no figures, no tables, *etc.*

I suggest avoiding the use of the defined acronyms in abstracts *i.e.*, spell them out rather than using the glossary commands. This is due to the fact that the `glossaries` package (that is being used to support acronyms) does not directly provide support for multiple languages and because I do not understand how to programmatically create plurals of acronyms in Swedish or other languages. Even in an English abstract, it is desirable to avoid using the glossary commands - as this makes subsequent processing of the abstracts harder - since one has to make sure that the list of acronyms and their definitions are provided to any program that will process this $\text{\LaTeX}$ source code. For this reason, later versions of this template include the `acronyms.tex` file after the metadata for DiVA.

C.6.2 Structure of the abstracts and summaries

The basic $\text{\LaTeX}$ structure for an abstract or summary is shown below (for the case of an English abstract and a Swedish summary *i.e.*, sammanfattning):

```
\begin{abstract}
  \markboth{\abstractname}{}
\begin{scontents}[store-env=lang]
eng
\end{scontents}

\begin{scontents}[store-env=abstracts,print-env=true]
here is where you abstract goes.
\end{scontents}

\subsection*{Keywords}
\begin{scontents}[store-env=keywords,print-env=true]
% If you set the EnglishKeywords earlier, you can
  ↳ retrieve them with:
\InsertKeywords{english}
% If you did not set the EnglishKeywords earlier then
  ↳ simply enter the comma separate keywords here:
%such as: Canvas Learning Management System, Docker
  ↳ containers, Performance tuning
\end{scontents}
\end{abstract}
```

```

\cleardoublepage
\babelpolyLangStart{swedish}
\begin{abstract}
  \markboth{\abstractname}{}
\begin{scontents}[store-env=lang]
swe
\end{scontents}
\begin{scontents}[store-env=abstracts,print-env=true]
Swedish summary goes here
\end{scontents}
\subsection*{Nyckelord}
\begin{scontents}[store-env=keywords,print-env=true]
% SwedishKeywords were set earlier, hence we can use
  ↪ alternative 2
\InsertKeywords{swedish}
\end{scontents}
\end{abstract}
\babelpolyLangStop{swedish}

```

It is important to note that the contents of the `scontents` environment for the abstracts are stored **verbatim**, *i.e.*, the $\LaTeX$ is **not** executed. The reason for this is to be able to later have a program that can manipulate the source $\LaTeX$ to convert it to HTML for use in announcements, calendar events, and for DiVA. This means that if you write the following:

```

\begin{scontents}[store-env=abstracts,print-env=true]
\input{abstract.txt}
\end{scontents}

```

what will end up in your abstract in the metadata save for DiVA will simply be: `"\input{abstract.tex}"` – which means that someone will have to cut and paste your actual abstract to insert it into DiVA.

It is also important to note that the following lines:

```

\begin{scontents}[store-env=lang]
eng
\end{scontents}

```

must be before the `scontents` environment for the abstracts and keywords – as these lines indicate what language the subsequent abstract and keywords are in. The three-character code used for the language is the ISO 639-2 Code – specifically the "B" (bibliographic) variant of these codes — as these codes are used in the DiVA metadata to tag what language is used.

C.6.3 Abstracts must be able to stand alone

The abstract needs to be able to stand alone; therefore, you **cannot** include citations to your references – as the references are **not** part of the abstract! It is possible (but very rare) to have footnotes as part of the abstract. However, you should be aware that quite often, if the abstract is manually entered in DiVA, the footnote might not be entered. In this case, unless your full text is available (*e.g.*, via DiVA), a reader might not have an easy way to find out what the footnote says.

C.6.4 Acronyms

You may want to define an acronym to help you with your writing, as this can both reduce the amount of typing and help your reader by providing consistent use of acronyms. The acronyms' definitions can be found in the file *lib/acronyms.tex*. The file contains some examples. I generally try to sort the lines to help find which acronyms I already have defined and keep track of the new one(s) I want to add.

C.6.5 Some predefined macros to help when writing

The file *lib/defines.tex* includes some macros that will help you when writing. This includes `\etc`, to give you “*etc.*”, `\eg`, `\ie`, and `\etal`. The file also defines `\first`, `\Second`, ... `\eighth` to give you *(i)*, *(ii)*, *(iii)*, ... *(viii)*. Note that ‘Second’ is written with an initial capital letter to avoid conflict with the unit ‘second’ in the `siunitx` package.

C.6.6 Additional abstract(s)

All theses at KTH are **required** to have an abstract in both *English* and *Swedish*. However, in addition to this, many students want to add abstracts in additional languages. The template comes pre-configured with places for abstracts in several other languages. If there is a language that you want to use that is not already supported, there are directions for how to add an additional language. If there are abstracts in languages that you do not want, please delete them or comment them out (see Section C.6.7).

C.6.7 Removing and hiding parts that you do not want

It is quite likely that you will find parts of the template that you do not want/need. One way of dealing with this is to delete them, and another way is to comment them out. Personally, I like to comment things out, in case I actually do want to be able to read it in the L^AT_EX file or uncomment it later. To comment out a portion of the file, simply use the following environment:

```
\begin{comment}
    *** what you want to comment out ***
\end{comment}
```

For example, if you are not interested in the Swedish language `todo` notes, you can look for lines with “\sweExpl” in them and comment them out (or delete them).

C.6.8 Removing the README_notes

At some point you will no longer want this README information. You can remove it by removing the line `\include{README_notes/README_notes}` – from the *examplethesis.tex* file. You can then remove the **README_notes** directory.

Unless you are an examiner or an administrator you can delete the file: `README_notes/README_examiner_notes.tex` and delete the include of this file from near the end of the template (*i.e.*, *examplethesis.tex*). You can also delete the directory **README_notes/README_examiner-figures**.

C.7 Copyright or Creative Commons License

It is possible to have several variants of the bookinfo page*:

copyright If you want to have a bookinfo page, include the line saying `\bookinfo`.

Creative Commons (CC) If you want to have a bookinfo page but want to have a Creative Commons license, then include `\bookinfo` and use and configure the `doclicense` package as described below.

*When printed double sided, the bookinfo page is the back of the title page.

none If you do **not** want to have a bookinfo page, comment the line saying \bookinfo page and add a \cleardoublepage.

For background about Creative Commons licenses, see: <https://www.kb.se/samverkan-och-utveckling/oppen-tillgang-och-bibsamkonsortiet/open-access-and-bibsam-consortium/open-access/creative-commons-faq-for-researchers.html> and <https://kib.ki.se/en/publish-analyse/publish-your-article-open-access/open-licence-your-publication-cc>.

Note that the lowercase version of the Creative Commons license has to be used in the modifier, *i.e.*, one of: by, by-nc, by-nd, by-nc-nd, by-sa, by-nc-sa, or zero. For the list of supported licenses, see the documentation for the doclicense package.

Note that if the doclicense package is used, it automatically redefines \bookinfo page to be \bookinfo pageCC.

C.7.1 Example configuration to have a CC BY-NC-ND license

```
\usepackage[
  type={CC},
  modifier={by-nc-nd},
  version={4.0},
  hyphenation={RaggedRight},
]{doclicense}
```

Note that the option “hyphenation=RaggedRight” can be used with the configuration of the package to set the license information with a ragged right margin rather than as a filled and justified paragraph.

C.7.2 Example configuration to have a CC BY-NC-ND license with a Euro symbol rather than a Dollar sign

```
\usepackage[
  type={CC},
  modifier={by-nc-nd},
```

```

version={4.0},
imagemodifier={-eu-88x31}, % to get Euro symbol rather
    ↪ than Dollar sign
hyphenation={RaggedRight},
]{doclicense}

```

C.7.3 Example configuration to have a CC0 license

```

\usepackage[
    type={CC},
    modifier={zero},
    version={1.0},
]{doclicense}

```

C.8 Use of fonts within the thesis

The choice of fonts is a very individual matter and may be affected by the kind of content that you are trying to write, the language that you are writing in, and what you want to convey to your reader. However, some points to keep in mind are:

- Use fonts with serifs for the body of your thesis, their presence makes it much easier for your reader.
- Use sans serif fonts for headings. This helps your reader distinguish them from the body.
- Be very careful when using fonts that are not widely available*. Unless you embed the fonts that you have used, your readers may not see what you want them to see. Ideally, you should embed all fonts – even if you only embed the subset you use.
- Although there are fonts that have a huge number of characters in them, they might not have the characters that you need.

*For example, even though it is widely used, not everyone has the Arial font. Additionally, it is a proprietary font; thus, you need to have an appropriate license to use it.

- There are also fonts that, although they have a vast number of characters in them, do not have the math table that $\text{\LaTeX}$ needs to be able to set mathematical content*.
- Many fonts are proprietary, thus you need to consider whether you have an appropriate license to use them.

What can you do when the fonts you use are missing characters that you need to use? One solution is to use a font that has the character(s) that you want and then make use of them in the places that you need to.

The details of working with different fonts and characters is a rather complex area and not for the faint-hearted. However, if you **really** want to have specific characters, $\text{\XeLaTeX}$ and $\text{\LuaTeX}$ have the means to help you realize what you want.

C.9 One big thesis file or a master file with includes of the parts

While many students split their thesis into multiple files (such as `introduction.tex`, `background.tex`, `method.tex`, `what-you-did.tex`, `results-and-analysis.tex`, `discussion.tex`, `conclusion-and-future-work.tex`) and then include these in their main document (with a series of `\includexxxx`), my experience is that it actually makes it hard to be consistent in the thesis. For example, you cannot do a simple global replacement when you have decided to introduce a particular acronym. It also makes searching for things difficult, as Overleaf's search function only works on individual files[†]. Personally, I find it hard to correct LaTeX errors when the file is split in this way - since Overleaf does not make it simple to find the root cause of a problem when the chapters are included in this way. There can be a problem with compiling the project in Overleaf, as Overleaf does not always handle the separate files as one document (unless you use the functions to tell LaTeX that a file is part of a larger document and identify the parent document). Although I have had some students do this splitting successfully and they liked being able to compile just

*An example of such a font is Google's Noto font. Even though it includes a vast number of characters, it lacks a math table – although there is an awareness of this missing feature

[†]Note that this is not a problem if you use emacs and a tags file, as this can do searching over the whole set of files and even a tags based query and replace.

a part of their report; I've personally had strange errors occur with it - hence I did not use this with the template.

There are some advantages to splitting the document into different parts:

- Overleaf has a limit on the number of changes that it can track - but this limit is per file! [Yes, I have gotten bitten by this when I have put in more changes and comments than the limit and had to stop marking up a manuscript.]
- Additionally, Overleaf has a per file size limit (*i.e.*, how large a file can be) - again this is per file [Yes, I have gotten bitten by this when exporting a Jupyter notebook that produced a LaTeX file larger than 50 MB.]
- This is useful when different students (in a 1st cycle degree project) are writing different parts of the report (in this case the divisions can be even at the section or subsection level).

Similarly, many students like to group their figures along with their chapters, *i.e.*, introducing a folder for each part of the report and placing both the text and the figures relevant to this section into the relevant folder. A similar approach can be used with included code snippets, tables, *etc.*

Ultimately, I think the main issue is the degree to which the separate files are separate and can be worked on as if they were very independent. This generally is true in third-cycle theses, as the chapters tend to be rather independent - typically with one conference/journal paper as the focus of a chapter. However, my experience is that first-cycle theses have very highly interdependent parts, while 2nd cycle theses are split between highly interdependent and highly independent.

However, some might find the question of splitting or not to be a matter of taste or perhaps different ways to approach organizing their writing. So you and your supervisor(s) and examiners might want to discuss what choice is most suitable for your purposes.

Appendix D

README and notes about the template

This document, written by Gerald Q. Maguire Jr, describes the thesis template that I have developed for use at **KTH Royal Institute of Technology (KTH)** and provides some background about why it is the way that it is. It is important to note that the template is **not prescriptive**, as not every thesis will have all the parts the template shows. However, if there is something that you decide to leave out, you should make a conscious decision to do so, and you should consider the impact this may have on your thesis being approved by the examiner.

Fundamental to the design of the template are several key factors:

- Helping students be successful in their degree project,
- Helping students produce a high-quality thesis, and
- Supporting all of the (relevant) phases of the degree project process.

Several thousand theses are written each year by **KTH** students. Every approved thesis will be entered into **Digitala Vetenskapliga Arkivet (DiVA)** (independent of whether the full text is made available via **DiVA**). Collecting the data necessary for **DiVA** was a major driving force in the design of the template. This data is useful for many of the phases of the degree project, such as announcing the oral presentation.

This template is **not** designed for use by **Interactive Media Technology (TIMTM)** and **Media Management (TMMTM)** students - as students in these two programs are using a different structure for their reports (there is another template available for them).

This document is a work in progress.

D.1 Introduction

This template evolved (radically) from an earlier thesis template that was widely used at **KTH**. The direction of this evolution was based on the DOCX template developed over many years for use with students for whom I was the examiner and/or supervisor. The suggested structure and contents of the thesis reflect my experience as an examiner for more than 600 degree projects and the experience I have had as a teacher and examiner for the course *II2202 Research Methodology and Scientific Writing*. The template also reflects my interest as a member of KTH's Language Committee in facilitating the parallel use of English and Swedish at **KTH**, as well as supporting other languages. The latter aspect reflects my experience with double-degree students, who often need to have at least the abstract of their thesis in their home university's language(s). The thesis template also reflects my experience in entering the metadata for hundreds of theses into **DiVA** and announcing a very large number of degree project seminars.

Section **D.4** describes several different groups of users and how the template is relevant to them.

Several major thoughts have influenced the design of this template:

- Thought 1** The template should help a student be successful in their degree project and help them produce a high-quality thesis in conjunction with their degree project.
- Thought 2** The template should help support all of the (relevant) phases of the degree project process.
- Thought 3** Redundant data entry should be minimized to increase consistency.
- Thought 4** There are several thousand theses written each year at **KTH**. Theses are the second most common type of publication at **KTH**.
- Thought 5** Every approved thesis will have at least its metadata entered into **DiVA**. **DiVA** features multi-language support for title, subtitle, abstract, and keywords.

D.2 Delimitations

This template is **not** designed for use by **TIMTM** and Media Management **TMMTM** students - as students in these two programs are using a different

structure for their reports (there is another template available for them).

Additionally, I have been told by one of my colleagues in applied mathematics that theses in this area generally do not follow the **Introduction, Methods, Results, and Discussion** (IMRAD) structure.

Some parts of the template are conditional based on the value of a switch: `\ifinswedish`. The idea is to easily have a single template that supports theses written in English or Swedish. However, in many places, the conditional has not been used but could be. Examples of this include the Swedish names for chapters and sections. Generally, this information is in a note after the English chapter or section name. More complete implementation of the use of this condition remains as future work.

The template does not fully support the G5 paper format. In particular, the **KTH** cover (produced with `\kthcover`) and back cover (produced with `\kthbackcover`) have only been adapted for A4 paper. Support for G5 paper remains as future work.

The handling of the subject area (Swedish: Område för examensarbete) is currently incomplete and remains as future work. Personally, I'm still struggling to understand the rules and how one knows what the correct values are (especially for cases of (i) dual degrees and (ii) combinations of technical subjects and education degrees).

D.3 Structure of the files for the template

Table D.1 shows the structure of the files for the template. These files are generally taken from an existing Overleaf project, a ZIP file, or a github.

One hope is that by automatically extracting information from various sources, this information is more likely to be *correct* and *consistent* (supporting **Thought 3**). This approach has been used to generate two of the files used for the template. These files are:

1. The file `custom_configuration.tex` contains macros and values for configuring a project. These values are generally expected to be known at the start of the project, *e.g.*, author(s), supervisor(s), examiner, course code for the degree project program code, *etc.* While this file can be manually edited, it was designed to be generated by a program that I have written that extracts most of the data from the Canvas course being used in conjunction with the degree project. One of the goals of using such a program is to extract data from Canvas automatically, the **KTH** profile **API**, **Kurs- och programplaneringssystemet (KOPPS)**, and

other sources. The macros for defining this information are described in Sections C.4, D.5.1, and D.5.2 - for authors, supervisors, and examiner (respectively).

2. The file `schools_and_programs.ins` contains the English and Swedish names of schools and programs. A program extracted this information from **KOPPS**.

We will assume that these files have been generated by someone. Later, we will examine who this someone might be for each of these files.

Table D.1: Structure of files for the template

bibstyle	directory containing files related to the style of the bibliography	
	<code>myIEEEtran.bst</code>	a bibtex style file
figures	directory containing files for figures	
kth	directory containing various files for the kththesis class	
	<code>kth-commands.tex</code>	definition of commands
	<code>kth-fonts.tex</code>	fonts for the template
	<code>kth-layout.tex</code>	the graphical layout
	<code>kth-packages.tex</code>	the packages used in the template
	<code>kth-sanity-check.tex</code>	some code to do sanity checking
lib	directory containing various library files	
	<code>acronyms.tex</code>	a place to define the acronyms that will or might be used
	<code>defines.tex</code>	some generally useful defines
	<code>includes-after-hyperref.tex</code>	a special include file for packages that have to be included after the hyperref package
	<code>includes.tex</code>	a centralized place to include packages that might be useful
	<code>kthcolors.tex</code>	defines a number of colors from the KTH palette
	<code>pdf_related_includes.tex</code>	includes to be able to add the title and other information to the PDF file
	<code>schools_and_programs.ins</code>	English and Swedish names of schools and the programs
<code>custom_configuration.tex</code>	macros and values for configuring a project	
<code>examplethesis.tex</code>	an example of the thesis itself	
<code>kth_logo.png</code>	the KTH logo for use on the cover	
<code>KTH_ROYAL_INSTITUTE_OF_TECHNOLOGY_logotype.png</code>	KTH logotype for use on the English language cover	
<code>kththesis.cls</code>	the kththesis class file	
<code>README_notes.tex</code>	these notes	
<code>references.bib</code>	references that may be cited in the thesis	

D.4 Expected users and their differences

This template is relevant to several different sets of users:

- Users 1** Author or Authors (see Section [C.1](#)),
- Users 2** Those working together with the author(s) during the degree project process (see Section [D.5](#)),
- Users 3** Administrative staff working with the document after it has been approved by the examiner (see Section [D.6](#)), and
- Users 4** The (hopefully) many (human) readers of the final document (see Section [D.7](#)).
- Users 5** The (hopefully) many computers reading the metadata and the full text of the final document (see Section [D.7.1](#)).
- Users 6** Those who are maintaining or updating this template (see Section [D.7.2](#)).

Each of these different sets of users has different needs and perspectives. The following subsections describe these needs and perspectives.

For information for authors, see Chapter [C](#) - located in the file `README_author.tex`.

D.5 Those working in parallel with the authors(s) during the degree project

Those working together with the author(s) during the degree project process include the examiner, supervisor(s), and the opponent(s).

D.5.1 Supervisor

If a degree project is done in industry, there is generally an industrial supervisor in addition to the academic supervisor(s). The template supports up to 5 supervisors (typically an academic supervisor, an industrial supervisor, and sometimes an additional academic or industrial supervisor). The choice of up to five reflects my experience, observation of prior theses in [DiVA](#), and a request by a student with four supervisors. Note that there is expected to be at least one supervisor. The supervisors are enumerated as A, B, C, D, and E.

For each of A, B, C, D, and E as appropriate, replace the "X" in the following macros:

`\supervisorXsLastname{ }` the last name of the supervisor
`\supervisorXsFirstname{ }` the first name of the supervisor
`\supervisorXsEmail{ }` e-mail address of the supervisor

If the supervisor is from within **KTH**, then add their KTHID, School, and Department info:

`\supervisorXsKTHID{ }` the supervisor's kthid
`\supervisorXsSchool{ }` the school of the supervisor
`\supervisorXsDepartment{ }` the department of the supervisor

If the supervisor is from outside of **KTH**, then add their organization with:

`\supervisorXsOrganization{ }` the supervisor's organization

D.5.2 Examiner

I assume that there is only a single examiner for a given thesis*. For this examiner, the relevant macros are:

`\examinersLastname{ }` the last name of the examiner
`\examinersFirstname{ }` the first name of the examiner
`\examinersEmail{ }` e-mail address of the examiner

If the examiner is from within **KTH**, then add their KTHID, School, and Department info:

`\examinersKTHID{ }` the examiner's kthid
`\examinersSchool{ }` the school of the examiner

*Statistically, there are very few theses with multiple examiners, and this generally occurs for students either in a double degree program or when there are two students in a 1st cycle degree project from different schools, then there might be one examiner for each student. As the case of more than one examiner occurs very infrequently, I have left it for future work. The pseudo-JSON structure is set up to handle multiple examiners, but additional macros would be needed in a similar fashion as used for multiple supervisors, and this metadata would have to be conditionally added where appropriate.

`\examinersDepartment{ }` the department of the examiner

If the examiner is from outside of **KTH**, then add their organization with:

`\examinersOrganization{ }` the examiner's organization

I assume that someone (such as the examiner) will generate the file: `custom_configuration.tex`. This assumption is based upon the fact that the examiner knows who the student or students are who will be working on a given degree project, who the supervisor or supervisors are, what program the student is in, course code, Ideally, this file should be generated automatically by some computer program so that each student or pair of students in a group gets a customized template automatically via the Canvas course. However, currently, the file is generated using a command line program (`create_customized_JSON_file.py`) to generate a **JavaScript Object Notation (JSON)** file. Subsequently, a separate program (`customize_LaTeX_project.py`) takes this **JSON** data and creates the appropriate **L^AT_EX** commands and inserts this information into the file and then inserts this file into a ZIP file, either replacing or augmenting the `custom_configuration.tex` within this ZIP file (if one exists). There is an option for this second program `-initialize` that causes the program to simply replace the file rather than appending the new information to the end of the file.

The above programs are available from <https://github.com/gqmaquirejr/E-learning>. The README file for this GitHub contains information about how to run the programs, their options, and gives examples.

D.5.3 Opponent(s) and oral presentation

Unlike the supervisors and examiner, the macros related to the opponent and oral presentation are in the `examplethesis.tex` file. The macro for the opponent(s) is:

`\opponentsNames{ }` the names (in normal name order) of
the opponent or opponents

When there are multiple opponents, separate their names with `'\&'`; for example, A. B. Normal `\&` A. X. E. Normal.

For the oral presentation, the following macros are filled in once the examiner has scheduled your oral presentation:

<code>\presentationDateAndTimeISO{}</code>	date and time of the presentation is ISO format, for example: 2022-03-15 13:00
<code>\presentationLanguage{}</code>	three letter abbreviation for the language of the presentation according to three letter ISO 639-2 Code – specifically the "B" (bibliographic) variant of these codes (note that this is the same language code used in DiVA), generally eng or swe
<code>\presentationRoom{}</code>	a room name and/or "via Zoom https://kth-se.zoom.us/j/ddddddddd "
<code>\presentationAddress{}</code>	location of the room, for example: Isafjordsgatan 22 (Kistagången 16)
<code>\presentationCity{}</code>	city where the presentation occurs, generally: Stockholm

D.6 Administrative staff

Once a thesis is approved by the examiner we need to add the TRITA number. The TRITA number is assigned by the student affairs office of the school from an annual series of numbers.

D.6.1 What is a TRITA number and why does each approved thesis get assigned one?

TRITA stands for Transactions for the Royal Institute of Technology, with the letter 'A' appended to it. The TRITA definition is the 1971 report, "Mall för publikationsserier vid Kungl. Tekniska högskolan i Stockholm", TRITA-LIB-1001, <http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-127656>.

The format for TRITA numbers for degree projects is TRITA-⟨school acronym⟩-EX-YYYY:nnnn, where nnnn is a sequential number starting from 1 each year with the numbers assigned in chronological order to approved theses (“numren delas ut kronologiskt först när examinatorn godkänt arbetet.” - according to one of KTH’s archivists). Note that the list of assigned TRITA numbers is archived each year*. The year, YYYY, is based on the year that the thesis was *approved*.

The TRITA number value can be set with a macro that takes two arguments: series and year:number as shown below:

```
% for entering the TRITA number for a thesis
\trita{TRITA-XXX-EX}{2022:00}
```

D.6.2 Where does the TRITA number go?

The TRITA number will appear on the back cover of the thesis. It is also stored as part of the metadata entered into **DiVA**.

D.6.3 What does this mean in practice?

Currently, at School of XXX[†] the TRITA number is only assigned to the thesis when the examiner has approved the thesis and submitted the PDF of the approved thesis (with cover) to the student affairs office. Of course, this does not make much sense because the back cover is already on the thesis! This means that someone in the student affairs office must either (i) edit the sequential number part of the TRITA number (using some PDF tool) or (ii) they need to make a new back cover and replace the existing back cover. A better solution would be to inform the examiner of the TRITA number and the examiner can see that this number is inserted into the macro shown above and this can enable the number to appear on the back cover and as an added bonus be included in the metadata for **DiVA**.

Note that it is expected that in 2023, this process will change – thus the assignment of the TRITA number and the application of the back cover would be done by the student affairs office (as only they have the relevant information)[‡].

*It seems that this archiving is done twice a year.

[†]Name replaced to neutralize the template.

[‡]Note to maintainers: This means that the back cover can be removed from this template.

D.6.4 Entering the metadata into DiVA

If a thesis has used this template the “For DIVA” page contains the metadata for **DiVA** and an administrator can cut and paste this data into **DiVA**. Alternatively, this metadata can be extracted with a program from the PDF file to produce a **JSON** file that can subsequently be used to create a MODS file for import into **DiVA**. The $\text{\LaTeX}$ compiler can in many cases produce a file called “fordiva.json” that contains the metadata.

The programs that can be used to extract data and to take a **JSON** file and create a MODS file are available from <https://github.com/gqmagui/rejr/E-learning>.

Note that the import of the MODS file does **not import the collaboration data**, even though this is in the file. This is a limitation of the **DiVA** import function. Therefore, this information has to be manually entered along with uploading of the PDF file itself.

D.7 (Human) Readers of the thesis

Some theses have very few downloads from **DiVA**, while some have had hundreds of thousands of downloads. Therefore, you should remember that you have a wide range of human readers of your thesis. The readers include other students looking for information related to their own thesis or because they are interested in the future work that you have suggested to work on for their own degree project. Additionally, researchers who are looking for your results may find your thesis relevant to them. In many cases, companies will look at theses for ideas about what the state of the art is - in several cases, theses have been important as “prior art” and this invalidated patents that had been issued if the patent was submitted after the thesis became public (hence it pays to get theses public as soon as possible). Other human readers are the UKÄ review teams that examine the degree programs offered at **KTH**. Finally, as **KTH** is a public agency, it is important that the general public know what is done at **KTH**^{*}.

D.7.1 Machines reading the metadata or full text of the thesis

The file `pdf_related_includes.tex` contains $\text{\LaTeX}$ code that stores the title, author(s), and keyword information into the PDF document in such

^{*}This is an important part of the Swedish Offentlighetsprincipen.

a way that if you ask for the properties of the PDF file you will get this data. This information makes it easier for machines to get this information from the PDF file.

Additionally, many search engines (such as Google’s search engine) mine **DiVA** for the metadata and if the full text of the thesis is published via **DiVA** then they also process the full text of the thesis. The result is that search engines can find the content in these theses. This is likely to increase the probability that someone will download your thesis if they think it is relevant to them – increasing the number of your human readers (see Section **D.7**).

D.7.2 Template author and maintainers

KTH periodically changes the cover design for theses, introduces new programs of study, eliminates programs of study, reorganizes administratively, and faculty move between schools, departments, and divisions. It can be expected that this template will need to evolve with these changes.

For example, if there is a change in schools or programs then there needs to be changes made to the file `schools_and_programs.ins`. While the current file was extracted from **KOPPS**, the program that does this will need to be replaced because further development of **KOPPS** has been terminated by KTH’s central IT unit which plans to transition all of this information to **Lokalt adb-baserat dokumentationssystemt (LADOK)**.

As another example, on 13 December 2021 there was a change in the **KTH** cover for 1st and 2nd theses, and the cover generator web service was shutdown. The initial draft version of the cover used a proprietary font (TheSans B4 SemiLight and TheSans B6 SemiBold). The version that was publicly introduced uses another proprietary font (Arial) and officially only existed as a DOCX file for a thesis in Swedish. The result is that I had to make my own version in $\text{\LaTeX}$ to try to emulate the DOCX cover. This led to a lot of effort, but one can get a reasonable cover with the correct font as described in Section **C.2**.

D.8 More about Abstracts

In addition to abstracts of the form shown in the template, there is another form of abstracts, called structured abstracts. These are described in Section **D.8.1**. Some automated checking of abstracts has been added to the template, see Section **D.8.2**.

D.8.1 Structured abstracts

In some fields, structured abstracts are commonly used. These typically have the following parts:

Background/Objectives

`<text>`

Method

`<text>`

Results

`<text>`

Conclusions

`<text>`

Such structured abstracts are typically longer (in word count) than typical abstracts. For an example of such a structured abstract, see [0]. The document is available in both English (*Humor as a protective factor against anxiety and depression*) and Spanish (*El humor como factor protector de la ansiedad y la depresión*) via <https://pmc.ncbi.nlm.nih.gov/articles/PMC6994741/>.

D.8.2 Automated checking of some properties of abstracts

To avoid common mistakes, the template now checks the contents of the abstracts and outputs warnings if the author puts a citation or cross-reference into the abstract.

The method of checking was inspired by the code used in the ACM Conference Proceedings templates to give a warning if the user uses a `\vspace` command or redefines `\baselinestretch`, see Listing D.1. To check the abstract, a pair of functions was introduced to save and restore commands, as was done in the package `phfnote`, see Listing D.2.

Listing D.1: Code snippet from ACM Conference Proceedings template

```

\let\@vspace@orig=\@vspace
\let\@vspacer@orig=\@vspacer
\apptocmd{\@vspace}{\ClassWarning{\@classname}{\string\
  ↪ vspace\space should
  only be used to provide space above/below surrounding
  objects}}{}{}
\apptocmd{\@vspacer}{\ClassWarning{\@classname}{\string\
  ↪ vspace\space should
  only be used to provide space above/below surrounding
  objects}}{}{}
\let\@vspace@acm=\@vspace
\let\@vspacer@acm=\@vspacer
\let\ACM@origbaselinestretch\baselinestretch
\AtEndDocument{\ifx\baselinestretch\ACM@origbaselinestretch\
  ↪ else
  \ClassError{\@classname}{An attempt to redefine
  \string\baselinestretch\space detected. Please do not do
  ↪ this for
  ACM submissions!}\fi}

```

Listing D.2: Code snippet from phfnote package

```

%% Macros from the package phfnote
% \phfnoteSaveDefs<{}>{identifier}<{}>{list of macro >names}
% ↪ saves the current definitions of the given list of
% ↪ macros under the identifier. The list of macros is
% ↪ specified as a comma-separated list of macro names.

\def\phfnoteSaveDefs#1#2{%
  \csgdef{phfnote@restoreddefs@#1}{}%
  \def\@tmpa{#2}%
  \for\next:=\@tmpa\do{%
    \global\csletcs{phfnote@restoreddefs@#1@\next}{\next}%
    \expandafter\xappto\csname phfnote@restoreddefs@#1\
      ↪ endcsname{%
      ↪ \noexpand\csletcs{\next}{phfnote@restoreddefs@#1@\next}
      ↪ %
    }%
  }%
}

% \phfnoteRestoreDef<{}>{identifier} - restores the saved
% ↪ macros
\def\phfnoteRestoreDefs#1{%
  \ifcsname phfnote@restoreddefs@#1\endcsname%
    \csname phfnote@restoreddefs@#1\endcsname%
  \else%
    \PackageError{phfnote}{\string\phfnoteRestoreDefs: no
      ↪ such
      definitions stored (#1)}{}
  \fi%
}

```

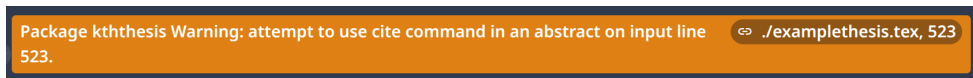
Given these two new macros, we use hooks (provided by the `etoolbox` package) to save the definition of `\cite` before the abstract and restore it afterwards – as well as to redefine `\cite` to produce a warning if used within the abstract. Note that in the PDF output, there will be **!?cite!?** at the place (in the PDF) where the `\cite` was - to draw the user’s attention to the error. The same mechanism is used for several other macros. The code for these hooks is shown in Listing D.3.

Listing D.3: Hooks to save and restore the cite and ref definitions before and after the abstract environment

```
\BeforeBeginEnvironment{abstract}{\phfnoteSaveDefs{
  ↪ origcmds}{cite, ref, Cref, cref}%
\renewcommand{\cite}[1]{\{\NotoSansFont \textcolor{red}{!}{\uuline{cite!}}\}\PackageWarning{kththesis}{attempt
  ↪ to use cite command in an abstract}}%
\renewcommand{\ref}[1]{\{\NotoSansFont \textcolor{red}{!}{\uuline{ref!}}\}\PackageWarning{kththesis}{attempt
  ↪ to use ref command in an abstract}}%
\renewcommand{\Cref}[1]{\{\NotoSansFont \textcolor{red}{!}{\uuline{Cref!}}\}\PackageWarning{kththesis}{
  ↪ attempt to use Cref command in an abstract}}%
\renewcommand{\cref}[1]{\{\NotoSansFont \textcolor{red}{!}{\uuline{cref!}}\}\PackageWarning{kththesis}{
  ↪ attempt to use cref command in an abstract}}%
}

\AfterEndEnvironment{abstract}{\phfnoteRestoreDefs{
  ↪ origcmds}}
```

The hooks shown in Listing D.3 produce a warning message that provides the page number where the `\cite` was used in an abstract (see Figure D.1). This code could be expanded to handle other undesirable commands in an abstract.



```
Package kththesis Warning: attempt to use cite command in an abstract on input line 523.
./examplethesis.tex, 523
```

Figure D.1: Example of a warning for use of a citation in an abstract)

The combination of two red Exclamation Question Marks (!?, unicode U+2049) with the command between them (double underlined) was selected as: (i) this string is unlikely in the user’s text, (ii) it was colored red to make it highly visible, and (iii) the offending command is double underlined to help make it visible (even to those who cannot see the color red). While this combination of characters is sometimes turned into an Interrobang glyph (‽, unicode U+203D), I have chosen to use the Exclamation Question Mark glyph as I found the Interrobang too subtle visually. Additionally, when using struck-through or struck-out characters rather than the double underline, it was too hard to read the name of the command. While the PDF does not include the arguments to the `\cite` or `\ref` command - these are in the source $\text{\LaTeX}$.

D.8.3 Tables and images in abstracts

This template and the tools that I have written to help enter the metadata into DiVA do **not** support tables and figures in abstracts.

As an abstract in DiVA can contain HTML code, it is technically possible to make tables and include figures in a DiVA abstract. The problems are (i) that I do not know a general means of transforming a table from LaTeX to HTML, and (ii) I do not know of a good way of passing the image (or images) that might be in a LaTeX abstract (via tools) to DiVA. The second problem might be addressed by attaching the image files to the PDF and adding code to the generated HTML to refer to these images. One problem with this is that the images would probably need to be uploaded as supplemental/supporting material, and then there would need to be some tool to update the abstract to refer to this uploaded file.

In a recent correction that I suggested for a thesis abstract in DiVA, I manually created the table in HTML to match the table that was in the abstract in the printed thesis.

The DiVA administrator who entered the thesis into DiVA had not put *any* of the abstract in, but simply put "See file" in the abstract field!

D.8.4 Graphical or visual abstracts

Karolinska Institutet (KI) has an excellent page about graphical abstracts, see [Graphical abstracts](#). At present, this template does **not** provide any special support for graphical abstracts. One can use the usual mechanisms to include a figure in the abstract; however, see Appendix [D.8.3](#).

I am unsure (i) how to deal with such figures with respect to facilitating their inclusion in DiVA (Could it be uploaded separately as supplemental material?), and (ii) it is unclear to me whether this graphical abstract would be placed before or after the keywords in the LaTeX file. This will require some further evaluation.

Recently, I helped create such a graphical abstract (for a journal paper) using `tikz`. This might have some advantages in terms of accessibility if the `tikz` source code were available to a (screen) reader.

D.9 Cover illustration

Some theses feature an optional cover illustration. Although I have not gotten an answer from the KTH communications unit about the specifications of the cover illustration, I have tried to infer something suitable from a DOCX file.

Due to the lack of a specification for the cover illustration, one should treat this feature of my template as an experimental feature that will be subject to change. Use at your own risk.

Two macros are introduced to make this process simpler, see Listing D.4. The first (`\coverIllustration`) provides the filename of the illustration to be used. The template includes an `image1.png` file. This blue rectangular image was taken from the DOCX template. The author should replace this file with the illustration that they want to use.

NB There are some constraints about what you can put on the cover. If in doubt, query KTH's [Communications Department \(COM\)](#).

The second macro (`\coverIllustrationCredit`) enables the author to acknowledge the creator of the illustration. Here, we will assume that the author has the creator's permission to include the illustration on the cover of their thesis. If the author has created their own illustration, then they do not need to specify any credit, as it is implicitly the work of the author.

NB The author may have to manually adjust the height and width options to the `\includegraphics` command in the `kththesis.cls` file to get the results that they want. If you make the figure too large, it will appear on a second page and **not** on the cover. If this occurs, you need to reduce the height of the illustration.

Listing D.4: Configuring a cover illustration

```
% It is possible to have a cover illustration
\coverIllustration{figures/image1.png}
% if so, the author might acknowledge this or give the
  ↳ copyright information for it with:
\coverIllustrationCredit{A. B. Normal}
```

D.10 While writing

As was noted in Section C.1, the thesis template contains lots of examples, notes, and comments. One method to provide additional information is the

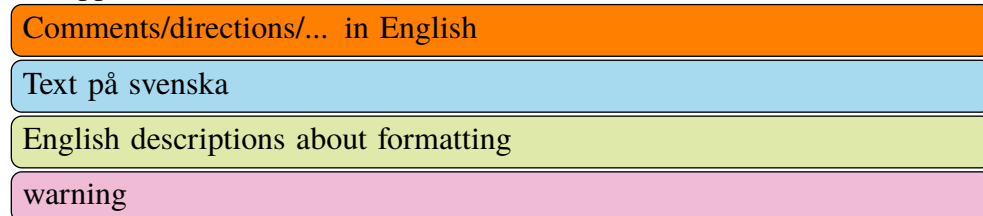
use of `\todo`. Several different types of `\todo` notes have been used in the thesis. These are described in Section [D.10.1](#).

D.10.1 Conventions for `\todo` notes

The example thesis text includes extensive comments, directions, and warnings. These follow the form shown below:

```
\generalExpl{Comments/directions/... in English}
\sweExpl{Text på svenska}
\engExpl{English descriptions about formatting}
\warningExpl{warning}
```

and appear as:



Each of the above is a macro, so as usual in $\text{\LaTeX}$ you can redefine it - even defining it to produce nothing! Several previous students have placed these re-definitions in the `custom_configuration.tex` file.

D.10.2 Turning on and off the `README_notes`

As the various `README` notes are targeted at different readers, you may or not want to see them. It is very easy to turn them on or off by adding or removing a percent (`'\%`') character before the relevant `\begin{comment}` and `\end{comment}` comments around each set of notes.

For example, if you are a student writing a thesis, I suggest turning off everything except for the `README_author.tex` and `README_notes.tex` sets of notes. However, I would suggest keeping the other `README` files around (at least for a little while) as a source of examples of how to do things. Despite having spent a very large number of hours working on the template and drafts of students' theses, I find some of the `README` files very helpful as a reminder of how to do things.

D.10.3 Removing the `README_notes`

At some point you will no longer want this `README` information. You can remove it by removing the line `\include{README_notes/README_notes}`

– from the `examplethesis.tex` file. If you have removed the other README* files from the **README_notes** directory, you can then remove the **README_notes** directory.

D.10.4 Removing the README_notes

At some point, you will no longer want this README information. You can remove it by removing the line `\include{README_notes/README_notes}` – from the `examplethesis.tex` file. If you have removed the other README* files from the **README_notes** directory, you can then remove the **README_notes** directory.

D.10.5 Removing unused fonts

This version of the template may also have some font information, in the form of OpenType Font files (with the extension “.otf”) and TrueType Font font files (with the extension “.ttf”). If you are not using these fonts (and no longer are using any of the README files), then you can delete these font files.

€€€€ For DIVA €€€€

```
{
  "Author1": { "Last name": "Wesslund",
    "First name": "Dante",
    "Local User Id": "u1eah7gh",
    "E-mail": "dantew@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
    }
  },
  "Cycle": "2",
  "Course code": "DA233X",
  "Credits": "30.0",
  "Degree1": { "Educational program": "Master's Programme, Machine Learning, 120 credits"
    , "programcode": "TMAIM"
    , "Degree": "Degree of Master of Science in Engineering"
    , "subjectArea": "Computer Science and Engineering"
  },
  "Title": {
    "Main title": "Adaptive Hybrid RAG",
    "Subtitle": "Optimizing the Efficiency-Accuracy Trade-off via Learned Query Routing",
    "Language": "eng"
  },
  "Alternative title": {
    "Main title": "Adaptiv hybrid-RAG",
    "Subtitle": "Optimering av avvägningen mellan effektivitet och noggrannhet med inlärld frågeroutning",
    "Language": "swe"
  },
  "Supervisor1": { "Last name": "Eriksson",
    "First name": "Arvid",
    "E-mail": "arveri@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
    }
  },
  "Supervisor2": { "Last name": "Zhang",
    "First name": "Peng",
    "Other organisation": "Ericsson"
  },
  "Examiner1": { "Last name": "Herman",
    "First name": "Pawel",
    "E-mail": "paherman@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
    }
  },
  "Cooperation": { "Partner_name": "Ericsson" },
  "National Subject Categories": "10210, 10208",
  "SDGs": "9, 12",
  "Other information": { "Year": "2026", "Number of pages": "xxl,91" },
  "Copyrightleft": "copyright",
  "Series": { },
  "Opponents": { "Name": "" },
  "Number of lang instances": "2",
  "Abstract[eng ]": €€€€

\generalExpl{Enter your abstract here!}
An abstract is (typically) about 250 and 350 words (1/2 A4-page) with the following components:
% key parts of the abstract
\begin{itemize}
  \item What is the topic area? (optional) Introduces the subject area for the project.
  \item Short problem statement
  \item Why was this problem worth a Bachelor's/'Masters thesis project? (\ie, why is the problem both significant and of a suitable degree of difficulty for a Bachelor's/'Masters thesis project? Why has no one else solved it yet?)
  \item How did you solve the problem? What was your method/insight?
  \item Results/Conclusions/Consequences/Impact: What are your key results/\linebreak[4]conclusions? What will others do based on your results? What can be done now that you have finished - that could not be done before your thesis project was completed?
\end{itemize}

"Abstract[swe ]": €€€€

\generalExpl{Enter your Swedish abstract or summary here!}

€€€€,
"Keywords[eng ]": €€€€
Retrieval-Augmented Generation, Graph RAG, Query routing, LightRAG, 2WikiMultihopQA, LLM-as-a-Judge €€€€,
€€€€,
"Keywords[swe ]": €€€€
Retrieval-Augmented Generation, grafbaserad RAG, frågeroutning, LightRAG, 2WikiMultihopQA, LLM-som-domare €€€€,
}
```



acronyms.tex

```
%%% Local Variables:
%%% mode: latex
%%% TeX-master: t
%%% End:
% The following command is used with glossaries-extra
\setabbreviationstyle[acronym]{long-short}
% The form of the entries in this file is \newacronym[label]{acronym}{phrase}
% or \newacronym[options]{label}{acronym}{phrase}
% see "User Manual for glossaries.sty" for the details about the options, one example is shown below
% note the specification of the long form plural in the line below
\newacronym[longplural={Debugging Information Entities}]{DIE}{DIE}{Debugging Information Entity}
%
% The following example also uses options
\newacronym[shortplural={OSes}, firstplural={operating systems (OSes)}]{OS}{OS}{operating system}

% note the use of a non-breaking dash in long text for the following acronym
\newacronym{IQL}{IQL}{Independent -QLearning}

% example of putting in a trademark on first expansion
\newacronym[first={NVIDIA OpenSHMEM Library (NVSHMEM\texttrademark)}]{NVSHMEM}{NVSHMEM}{NVIDIA OpenSHMEM
↔ Library}

\newacronym{API}{API}{Application Programming Interface}
\newacronym{auprc}{AUPRC}{area under the precision-recall curve}
\newacronym{bert}{BERT}{Bidirectional Encoder Representations from Transformers}
\newacronym{DiVA}{DiVA}{Digitala Vetenskapliga Arkivet}
\newacronym{gpu}{GPU}{Graphics Processing Unit}
\newacronym{IMRAD}{IMRAD}{Introduction, Methods, Results, and Discussion}
\newacronym{JSON}{JSON}{JavaScript Object Notation}
\newacronym{jsonl}{JSONL}{JSON Lines}
\newacronym{KI}{KI}{Karolinska Institutet}
\newacronym{KOPPS}{KOPPS}{Kurs- och programplaneringssystemet}
\newacronym{LADOK}{LADOK}{Lokalt -adbbaserat dokumentationssystem}
\newacronym[longplural={large language models}]{llm}{LLM}{large language model}
\newacronym{llmjude}{LLM-as-a-Judge}{large language model as a judge}
\newacronym{nlp}{NLP}{natural language processing}
\newacronym{qa}{QA}{question answering}
\newacronym{rag}{RAG}{Retrieval-Augmented Generation}
\newacronym{tfidf}{TF-IDF}{term frequency-inverse document frequency}
\newacronym{TIMTM}{TIMTM}{Interactive Media Technology}
\newacronym{TMMTM}{TMMTM}{Media Managemen}
\newacronym{KTH}{KTH}{KTH Royal Institute of Technology}

\newacronym{LAN}{LAN}{Local Area Network}
\newacronym{VM}{VM}{virtual machine}
% note the use of a non-breaking dash in the following acronym
\newacronym{WiFi}{-WiFi}{Wireless Fidelity}

\newacronym{WLAN}{WLAN}{Wireless Local Area Network}
\newacronym{UN}{UN}{United Nations}
\newacronym{SDG}{SDG}{Sustainable Development Goal}
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